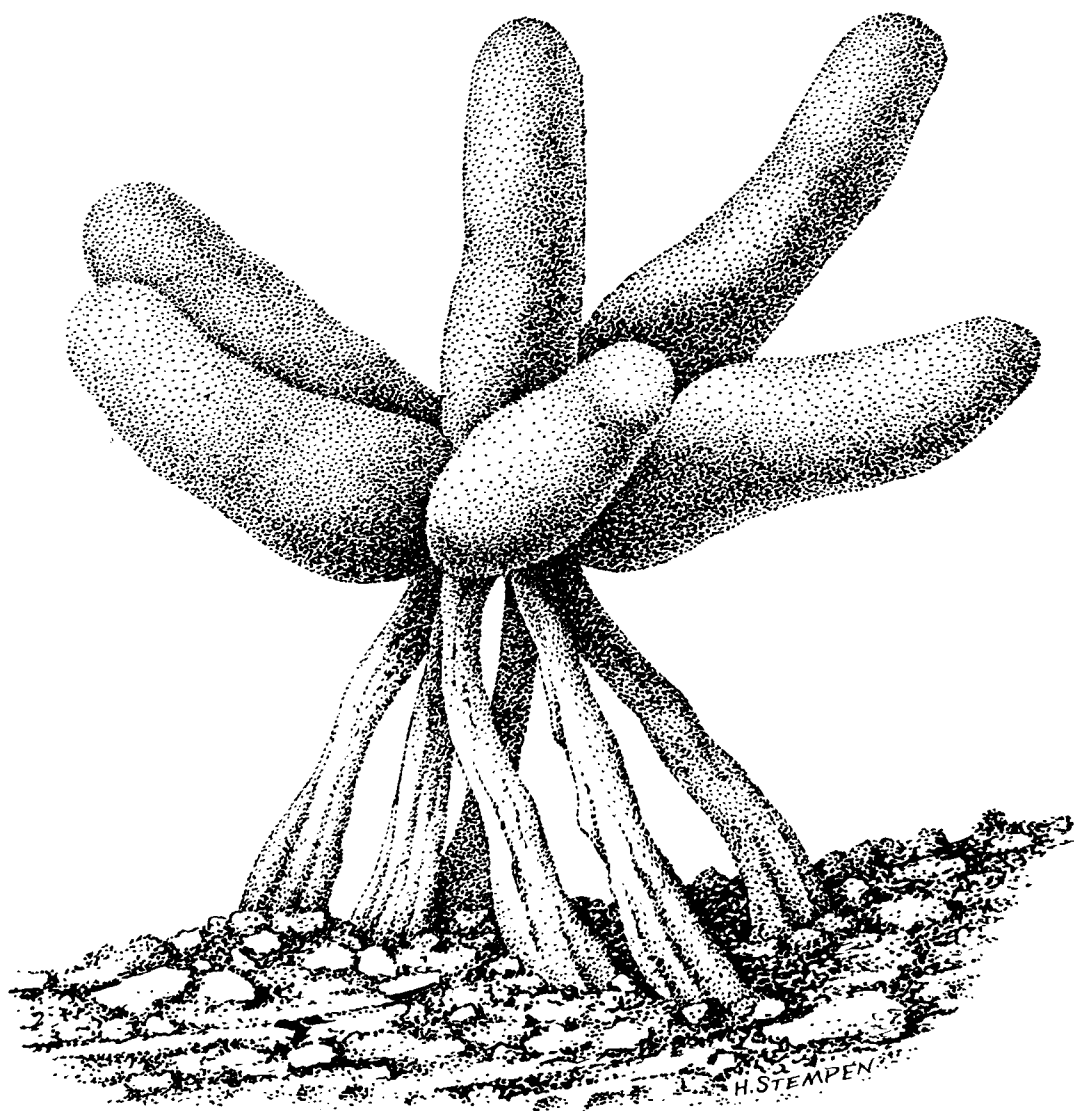


Mycological
Society of America
NEWSLETTER



vol. 30 no. 1 June ~~1978~~ 1979

SUSTAINING MEMBERS

ANALYTAB PRODUCTS
Division of Ayerst Laboratories
Plainview, New York 11803

AYERST LABORATORIES
Division of Ayerst, McKenna & Harrison Limited
1025 Laurentian Blvd., P. O. Box 6082
Montreal, Canada H3C 3A7

BELLCO GLASS, INC.
Complete offering of biological laboratory glassware & equipment
P. O. Box B, Vineland, New Jersey 08360

BUTLER COUNTY MUSHROOM FARM
West Winfield, Pennsylvania 16062

CALBIOCHEM
P. O. Box 12087
San Diego, California 92112

CAMPBELL SOUP COMPANY
Camden, New Jersey 08101

DIAMOND SHAMROCK CORPORATION
T. R. Evans Research Center
P. O. Box 348
Painesville, Ohio 44077

DIFCO LABORATORY PRODUCTS
The complete line of
microbiological reagents and media
P. O. Box 1058-A
Detroit, Michigan 48232

DUPONT COMPANY, INSTRUMENT PRODUCTS
Biomedical Division
Peck's Lane
Newtown, Connecticut 06470

FUNK SEEDS INTERNATIONAL
A Company of CIBA-GEIGY Corporation
Funk's G-Hybrid Corn, Sorghum and Farm Seeds
Bloomington, Illinois 61701

HOECHST-ROUSSEL PHARMACEUTICALS INC.
Route 202-206 North
Somerville, New Jersey 08876

HOFFMAN-LA ROCHE, INC.
Pharmaceuticals, vitamins and aromatic chemicals
Nutley, New Jersey 07110

LADD RESEARCH INDUSTRIES
P. O. Box 901
Burlington, Vermont 05401

LANE SCIENCE EQUIPMENT CO.
Complete line of museum storage cabinets
especially herbarium cabinets--airtight for permanent protection
105 Chambers St.
New York, New York 10007

ELI LILLY AND COMPANY
Pharmaceuticals, agricultural and industrial products and cosmetics
307 East McCarty, Indianapolis, Indiana 46206

MERCK SHARP AND DOHME RESEARCH LABORATORIES
Division of Merck & Co., Inc.
Rahway, New Jersey 07065

NEW BRUNSWICK SCIENTIFIC CO., INC.
Manufacturers of Precision Research Apparatus
P. O. Box 986, 44 Talmadge Road
Edison, New Jersey 08817

PFIZER, INC.
Fine chemicals and pharmaceuticals
by means of microorganisms
235 East 42nd Street
New York, New York 10017

PIONEER HI-BRED INTERNATIONAL, INCORPORATED
1206 Mulberry Street
Des Moines, Iowa 50308

GROUP QUALITY ASSURANCE
The Quaker Oats Company
John Stuart Research Laboratories
617 West Main Street
Barrington, Illinois 60010

ROHM AND HAAS COMPANY
Research Division
P. O. Box 219
Bristol, Pennsylvania 19007

SCHERING CORPORATION
Pharmaceutical manufacturers
Bloomfield, New Jersey 07003

SMITH KLINE & FRENCH LABORATORIES
Prescription medicines and other health care products
Division of Smith Kline Corporation
Philadelphia, Pennsylvania 19101

TRIARCH INCORPORATED
Quality prepared microscope slides, catalog listed
or custom prepared to your specifications
Ripon, Wisconsin 54971

THE UPJOHN COMPANY
Fine pharmaceuticals since 1886
Kalamazoo, Michigan 49001

WARNER-LAMBERT/PARKE-DAVIS
Pharmaceutical Research Division
Detroit, Michigan 48232

LETTER FROM THE EDITOR

As the new editor of the MSA newsletter I would like to express my hope that I can be of service to members of the society and can continue the tradition of high quality and information content established by my predecessors.

The format of this issue of the newsletter is essentially that used by the previous editor. If members of the society have suggestions about other departments they would like to see added to the newsletter, they should contact me. Don't feel obliged to limit contributions to the categories listed in the newsletter questionnaire.

The beautiful cover illustration of Arcyria cinerea was provided by Henry Stemper. In the future I would like to use examples of American mycological illustration in the cover design, including both historical material and contemporary drawings. Contributions to this department are solicited from the membership. Illustrations should be of publishable quality and scientific accuracy. Subjects other than mushrooms are particularly welcome.

I appologize for having mailed the newsletter questionnaires so late and regret any inconvenience this may have caused. Hopefully this situation will not occur again.

GENERAL ANNOUNCEMENTS

A proposal has been made to the Nomenclature Secretariat of the International Mycological Association that the 'Nomenclature of fossil fungi' is urgently in need of review. The Secretariat is, therefore, prepared to create a subcommittee to this effect and requests that all persons active in, or interested in, fossil fungi notify the chairman of their willingness to serve on the subcommittee so that they can be appointed members. (K. T. van Warmelo)

Der Pilz, a shop specializing in books, art objects, and accessories with a mushroom motif, has opened at Mittelweg 21, 2000 Hamburg 13, Germany. tel. 417597

The Council for Optical Radiation Measurements (CORM) has reorganized. For more information contact: Fred N. Billmeyer, Department of Chemistry, Rensselaer Polytechnic Institute, Troy, NY 12181.

The Marcel Dekker, Inc., invites Mycologists to submit proposals for manuscripts to be considered for publication in a newly formed Mycology series. The Series is broadly conceived as to scope and will publish textbooks and well as original research works and monographs. (Paul A. Lemke)

The Ad Hoc Committee on the Golden Anniversary of the MSA is soliciting ideas from the membership on appropriate ways to celebrate this special occasion at the 1981 AIBS meetings. (M. L. Farr, J. W. Kimbrough, J. D. Rogers, C. T. Rogerson)

Workers in marine Mycology who would like to receive the Aquatic Microbiology Newsletter, distributed to aquatic scientists since 1961, should contact S. P. Meyers.

SYMPOSIA, MEETINGS, FORAYS, CONFERENCES

The society for Industrial Microbiology will hold its 30th annual meeting August 12-17 at Carnegie-Mellon Institute, Pittsburgh, Pennsylvania. (Ann Kulback, 1401 Wilson Boulevard, Arlington, Va. 22209)

The annual lecturer at the MSA meeting in Stillwater will be Dr. I. B. Heath of York University in Ontario.

A Gordon Research Conference on Fungal Metabolism will be held at the Brewster Academy in Wolfeboro, New Hampshire, July 28- August 1, 1980. (Paul A. Lemke, Stuart Tanenbaum)

A paleomycological colloquium will be held at the Laboratoire de Micro-paleontologie, Paris, France, in July, 1980. (Marcel Loquin)

The Mycological Society of San Francisco is holding their Sierra Foray September 8-9, 1979, the Key Council Foray in Washington State October 13-14, 1979, the Point Reyes Foray on November 4, Gualala Foray on November 18, and SF watershed foray on December 8. They publish a calendar of other club activities.

The second International Symposia on Biology and Management of Mangroves and Tropical Shallow Water Communities will be held in Port Moresby, New Guinea, July 20-August 2, 1980. (David H. Montgomery, Biological Sciences Dept., California Polytechnic State University, San Luis Obispo, CA 93407)

The Beltsville Agricultural Research Center is sponsoring a symposium on Biological Control in Crop Production, May 18-21, 1980. (E. M. Dougherty, Publicity Committee, BARC Symposium V, Bld. 011A, Beltsville Agricultural Research Center-West, Beltsville, MD 20705)

The American Phytopathological Society is sponsoring a discussion session on Mycoparasites, to be held at their annual meeting August 9 at 7 pm in the Blue Room of the Shoreham Hotel, Washington, DC (APS Mycological Committee, Ruth Taber, Texas A & M. University)

SYMPOSIA, MEETINGS, FORAYS, CONFERENCES

The Association of Southeastern Biologists will be meeting at the University of South Florida in April, 1980. (Fred I. Eilens, Biology Department, University of South Florida, Tampa, FLA 33620)

The third annual Northeastern Mycological Foray will be held at Eastern Connecticut State College, August 24-26, 1979. (James Kronick)

The third International Marine Mycology Symposium will be held at the University of North Carolina Institute of Marine Sciences, Morehead City, North Carolina (no date given). (J. Kohlmeier)

The North American Mycological Association is holding a foray September 27-30, 1979, in Carroll County, Ohio. (Ohio Mushroom Society)(Harold Keller)

The A. H. Smith Lake States Foray will be held on September 21 and 22 in western Wisconsin near La Crosse (Allen G. Nelson)

NEW MYCOLOGICAL RESEARCH PROJECTS (see also addenda)

Dennis Gaunt is working on the serological identification of pathogenic systemic fungi by the gel immunodiffusion technique.

John Taylor is beginning a two year project on 'Mitosis and Systematics of the Basidiomycotina - - Dacrymycetaceae and Tulasnellaceae.'

James M. Kronick is investigating Automated identification of yeasts, primarily for medical diagnostic applications.

George E. Templeton is working on 'Biological control of weeds with fungal plant pathogens'. The project has been approved as Southern Regional Research Project S-136 by directors of Southern Agricultural Experiment Stations.

L. L. Burpee is continuing is continuing studies of Rhizoctonia species other than R. solani.

Edward Felix and Garry T. Cole are beginning a study on the effects of salinity on the growth, conidial development, spore germination, and ultrastructure of salt-tolerant aquatic fungi (in particular marine hyphomycetes occurring on sea grass: Thalassia testudineum)

Walter Kaiser is working on diseases that effect Phaseolus and Lens in the USDA Plant Introduction collection at Pullman, Washington, and will be screening the PI collection for resistance to important fungus diseases.

Garry T. Cole is investigating developmental and ultrastructural aspects of conidiomata in Coelomycetous fungi.

Roger E. Phillips is studying fungi associated with several intertidal red algae.

George Hatfield and Kenneth Cochran have initiated a joint research project on natural products with antiviral activity. (see also fungi wanted)

D. S. Chahal is studying bioconversion of lignocellulosic wastes into single-cell protein with Chaetomium cellulolyticum.

Mary Palm is investigating the taxonomy and biology of selected members of the Boletaceae in northern Minnesota (with E. L. Stuart)

Carlene Raper and several students at Wellesley College are investigating the genetic and cytological basis for the phenomenon of asymmetry in survival of de-dikaryotized monokaryons of Schizophyllum commune. They are employing the phenomenon of protoplast formation and regeneration to obtain large samples of monokaryons from dikaryons. They are also working on a long term project to develop bioassays

on the incompatibility-gene products through the use of protoplasts and protoplast fusion techniques.

FORTHCOMING COURSES IN MYCOLOGY

Medical Mycology - a survey course of the yeasts, molds, and actinomycetes most likely to be encountered by the medical mycologist in a diagnostic laboratory, will be offered fall quarter at the University of Georgia (G. E. Michaels)

A course in field identification of mushrooms will be taught by Fred van de Bogart at the Teton Science School, August 27-31, 1979. (Kurt Bademacher, Director, Teton Science School, Box 68, Kelly, WY 83011, 307-733-4765)

Barbara Dyko, of Michigan State University, will teach a course in isolation and identification of Phytopathogenic fungi, 28 June - 7 August 1979. Heavy emphasis will be given to Hyphomycetes and Coelomycetes.

Harvard University Extension offers evening classes in Mycology (Fall term; D. H. Pfister) and Lichenology (Spring term; M. A. Sherwood). (Commission on Extension Courses, 20 Garden St., Cambridge, MA 02138)

Rensselaer Polytechnic Institute offers a Summer program in color technology, June 4-8 and 11-15, 1979. (Richard J. Teich, Office of Continuing Studies, R.P.I., Troy, NY 12181)

R. Singer will teach a course in 'Grosspilze des Wiener Raumes' at the University of Vienna, Austria, August-September, 1979. (Botanisches Institut d. Universität Wien, Rennweg 14, A-1030, Wien, Austria)

FUNGI FOR DISTRIBUTION

Dermatophytes, pathogenic yeasts, and some agents of chromoblastomycosis (Dennis Gaunt)

Sorosphaera veronicae Schroet. on Veronica spp., preserved in FAA (George E. Templeton)

Cultures of Rhizoctonia spp. other than R. solani (L. L. Burpee)

Hypovirulent, virus-containing strains of Endothia parasitica - cultures - with plant quarantine permit (Sandra Agnostakis)

Martha M. Katter has information available on the hypogeous Ascomycetes and Basidiomycetes in Poderosa pine forests in the western United States and Canada - distribution records.

Nichole O'Neill has cultures available of Rhizoctonia spp. and Helminthosporium from turfgrass.

Amy Rossman has single spore isolates of Nectria species which produce Fusarium, Cylindrocarpon, Acremonium, and Cylindrocladium states, from the Azores and Madeira.

Stanley Flegler can supply Cyathus stercorearius cultures.

FUNGI NEEDED FOR RESEARCH PROJECTS

GENERAL

George Templeton would like fungi pathogenic on the aerial parts of cocklebur, morning glory, water hyacinth, hemp sesbania, sicklepod, johnson grass, prickly sida, nut sedge, and alligator weed.

Robert Fogel would like either cultures or specimens of hypogeous fungi.

MYXOMYCETES

Charles Miller would like sporangia, cysts, or cystosori of any Plasmodiophoraceous fungus.

Harold Keller needs specimens of Licea, Clastoderma, and Perichaena.

PHYCOMYCETES (see also addenda)

Chris Walker requests Endogonaceae for taxonomic research and for building up a reference collection. Glomus mossae and G. fasciculatus would be especially welcome.

ASCOMYCETES

Martha Sherwood solicits large collections of Phacidiales (in the broad sense) for an exsiccati.

Amy Rossman would like fresh specimens of Nectria and Calonectria spp. from tropical areas, and unidentified specimens of Nectria and Calonectria.

Sandra Agnostakis wants cultures or specimens of Endothia fluens (= E. radicalis) Walstine Steffens needs cultures of Eleutherascus spp. other than E. peruvianus.

J. D. Jensen needs cultures or recent (less than 1 year old) collections of Hypoxylon serpens.

George Orr would like Gymnoascaceae and related fungi.

A. Weintraub is looking for cultures of Morchella esculenta.

Paul Conant is still looking for FAA fixed material of Phyllactinia corylea with characteristic spear-like, bulbous based appendages.

Richard Korf wants any discomycetes collected in the Azores, Madeira, Canary Islands, or Cape Verde.

Hin Chung Wong would like any Monascus species.

Jim Kimbrough wants cultures or specimens of coprophilous discomycetes and termite-inhabiting fungi.

John Leslie would like cultures derived from natural populations of Neurospora.

BASIDIOMYCETES

John McCain wants specimens of rust genera that sporulate supra-stromatally (Hemileia, Gerwasia, Desmella, Edythea, Prosopodium) fresh or suitably preserved for electron microscopy.

Edmund Badham wants cultures of Basidiomycetes that can be fruited in culture.

Kenneth Cochran wants 5 gm (dry weight) samples, less than two years old, of taxonomically distinct and well characterized mushrooms, for his studies on antiviral compounds.

Fred van der Bogart needs specimens of Coprinus.

FUNGI IMPERFECTI

E. Felix and G. T. Cole would like cultures of marine deuteromycetes isolated from sea grasses and any salt-tolerant deuteromycete isolated from a saline (especially hypersaline) environment.

L. L. Burpee wants cultures of Rhizoctonia other than R. solani.

Nichole O'Neill is looking for cultures of Helminthosporium isolated from Gramineae.

IDENTIFICATION SERVICES

MYXOMYCETES

Harold Keller will identify specimens of Licea, Clastoderma, and Perichaena.

Charles Miller is willing to identify Plasmodiophoraceae.

PHYCOMYCETES (see also addenda)

Richard Humber will identify entomogenous fungi, especially Entomophthorales.

Chris Walker will identify Endogenaceae.

ASCOMYCETES

John Haines is willing to identify Hyaloscyphaceae.

Jim Kimbrough will identify coprophilous discomycetes and termite inhabiting fungi.

George Orr will identify any Gymnoascaceae except Nannizzia and Arthroderma by arrangement.

Amy Rossman is willing to identify Calonectria and Nectria specimens.

Martha Sherwood will identify specimens of Ostropales and Rhytismataceae.

BASIDIOMYCETES

The research committee of the North American Mycological Association is offering a macromycete identification service to members who are able to provide (1) field notes describing fresh material, (2) dried specimens, (3) a color photograph, (4) a spore print, and (5) a stamped, self-addressed card for reply. Coordinator for this service is Leo Tanghe, 852 Stone Road, Rochester, NY 14616.

Robert Fogel will identify hypogeous Basidiomycetes, including Leucogaster, Leucophleps, Hymenogaster, and Sclerogaster.

FUNGI IMPERFECTI

Mary Palm is willing to identify Trichocladium and closely related genera.

Jan Kohlmeyer is willing to identify higher marine filamentous fungi.

L. L. Burpee will identify Rhizoctonia spp. and fungi associated with Gramineae.

PUBLICATIONS AVAILABLE

* Sandra Agnostakis

The American Chestnut: new hope for a fallen giant. Ct. Agr. Ext. Bull. 777 (1978). Free

* Robert A. Samson, Centraalbureau voor Schimmelcultures

Booklet commemorating the 75th anniversary of the Centraalbureau voor Schimmelcultures. Free.

Studies in Mycology 17. Schipper, M. A. A. (1). On certain species of Mucor with a key to all the accepted species. (2). On the genera Rhizomucor and Parasitella. 70 pp. hfl 20,-

Studies in Mycology 18. Samson, R. A. A compilation of the Aspergilli described since 1965. 40 pp. hfl 10,-

* Bryce Kendrick

The whole fungus - the sexual-asexual synthesis. Proceedings of Kananaskis II. 2 vols, 793 p. \$ 16.00 + 2.00 postage and handling. Send payment with order. List of other University of Waterloo publications available.

* Moselio Schaechter, 855 Commonwealth Ave., Newton, MA 02159

Occurrence of New England Fungi as reported by members of the Boston Mycological Club, 1964-75. 16 p. price not given.

This organization has also published a number of other pamphlets of interest to the amateur mushroom collector. Write for details.

* Larry J. Littlefield and Michele Heath would like to call society member's attention to their recently published book, *Ultrastructure of Rust Fungi*. (Academic Press, 1979)

*John L. Maas

Kearney, T. H. & R. H. Peebles, 1942. Flowering plants and ferns of Arizona. 12.50

Bessey, E. A. (1950). Morphology and taxonomy of fungi. orig. ed. 15.00

Chupp, C. (1953). A monograph of *Cercospora*. 8.00

Kendrick, B. (1971). Taxonomy of Fungi Imperfecti. 12.00

Miller, O. K. and D. F. Farr (1975). An index of the common fungi in North America. 10.00

Taylor, R. (1897-98). Mushrooms of America nos 1-5 (with color plates). 10.00

Wehmeyer, L. E. (1975). The Pyrenomycetous fungi. *Mycologia* memoir 6. 10.00

Wollenweber, H. W. & G. Reinking (1927). Tropical *Fusaria*. 10.00

Thom and Raper (1945). A manual of the *Aspergilli*. 10.00

Mycologia vols. 54-69 (1962-1977) 6.00 per volume, vol. 39, nos 3, 4, 6, 3.00.

Please add .50 postage per item. Large list of other books and reprints on Fungi, plant diseases, Flowering plants, and Agricultural Yearbooks sent on request.

* Gaston Guzmán

Manzi, J. (1976). Hongos comestibles y venenosos. Mexico. 32 col. pl. 6.00

Tosco & Fanelli, 1973. Las setas. Barcelona (Color plates) 7.00

Guzman, 1978. Hongos. 186 color plates. 7.00

Klejn, 1962. Mushrooms and other fungi. Oldbourne. 8.00

Major, 1974. Mushrooms, toadstools, and fungi. 5.00

* Donald H. Pfister & Geraldine Kaye

The Farlow Reference Library will furnish on request a list of duplicate books and journals for sale, including many older phytopathological references and some useful taxonomic works. They will also supply lists of current publications for sale and reprints available for distribution. Many of the reprints date from the late 19th and early 20th century, and some are mycological classics.

* Rene Pomerleau

A complete collection of *Revue de Mycologie*, 42 volumes, 1936-1978. 1000 American dollars.

Peck's Annual Reports, 22-38 bound (1869-84); 39-48 unbound (1885-94); 1898-99 unbound, 1900-1911 bound, 1912-1924. 1500 American dollars.

* David Gottlieb

Mycologia, vols 43-71 (1951-79), missing vol. 66:1

Phytopathology vols. 31-79 (1941-79), missing vol. 36:7

Journal of Bacteriology, vol. 55-137 (1948-79), Missing vol. 109:2, 115:1

* Elwyn Reese

Lloyd, *Mycological Writings*, 1898-1925 in 7 vol. \$ 300

Smith and Sowerby, *English Botany, or colored figures of British Plants*, v.8-12, 1841-46, with 1172 colored plates. \$300 These volumes cover ferns, mosses, algae, and some fungi.

Greville. *Scottish Cryptogamic Flora*, v.1-6. 1823-38. \$400 (360 colored plates)

* Diane Wagner

15 copies of IMC-II abstracts, volume II only, are available. Send \$ 2.00 to cover postage and handling.

* Martha Sherwood

The Ostropalean Fungi. reprinted from Mycotaxon 5: 1-277, available in exchange for similar publications.

* Melvin S. Fuller

"Lower Fungi in the Laboratory" is still available. See Mycologia 71: 458 for review and details on purchase. The Department of Botany, University of Georgia, will publish a second book in August of 1979. It is "Zygomycetes in Culture", written and profusely illustrated by Kerry O'Donnel.

* G. T. Cole and R. A. Sampson would like to call attention to "Patterns of Development in Conidial Fungi," now in print, published by Pitman Publishing, London.

* John Haines

Sets of Peck's Reports are still available from the New York State Museum. Write for details.

* Richard Humber

World directory for current research on the entomopathogenic fungus Entomophthora. Free.

OTHER MISCELLANEOUS ITEMS FOR SALE

* A. Weintraub

Leitz monocular microscope w/mirror, mechanical stage, good cond. 225.00

Spencer A0 binocular microscope w/case, mech. stage, good cond. 200.00

Bausch & Lomb binoc. in case, lamp, mechanical stage, good cond. 385.00

Shipping charges extra.

VACANCIES FOR MYCOLOGISTS: POSTDOCTORAL POSITIONS AND REAL JOBS

A postdoctoral position is available in the Department of Biological Sciences, Purdue University, to study glycoprotein synthesis and export during papilla formation in *Blastocladiella*. Contact I. S. Lovett.

Dr. Roger Smith, of Forintek Canada Corporation, Vancouver, B. C., is looking for a wood products biodeterioration Mycologist to do applied research. Applicant should have a PhD and a good knowledge of wood-inhabiting fungi. Person with postdoctoral experience preferred.

Two Postdoctoral positions at the Boyce-Thompson Institute are available to support an established program studying the entomopathogenic fungi. (1) A fungal spore physiologist to investigate the mechanism of Entomophthora resting spore germination, and (2) a fungal geneticist to study basic genetics of Entomophthora to enhance field utility through fungicide resistance and increased pathogenicity. Salary 12,000 per year or higher depending on experience. The Biocontrol program at BTI has 21 staff members including 9 PhDs. Send resume, bibliography, and names of three referees to Dr. Richard Soper.

A postdoctoral position is open at the Erindale campus of the University of Toronto, in the general area of the developmental biology of Eukaryotic microbes. Applicant should have strong background in biochemistry and cell biology. Prior work with nucleic acids, nuclear protein, or nucleic acid polymerases is desirable. Funds available for two years. Send CV to P. A. Horgen.

FELLOWSHIPS AND ASSISTANTSHIPS AVAILABLE

A research assistantship is available in developmental Mycology at the University of Texas, Austin. Contact G. T. Cole

Persons interested in graduate study in Botany (Mycology) at the University of Georgia should contact Melvin S. Fuller.

Persons wishing to study fungal dynamics in ponds and fungi of Coral reefs should contact Bryce Kendrick. Scuba certification is desirable.

Several graduate assistantships are available for the study of Mycology leading to M.S. and PhD degrees at Auburn University, Alabama. Contact. G. Morgan-Jones.

Graduate assistantships in Botany are available at Ohio University, Athens, Ohio. Contact Charles E. Miller.

Assistantships are available in Mycology at the University of Florida, Gainesville. Contact J. W. Kimbrough.

A graduate research Assistantship for a masters degree candidate interested in working on freshwater fungi is available at Western Illinois University. Contact Robert V. Gessner for further information.

MYCOLOGISTS LOOKING FOR EMPLOYMENT

Louis S. Tisa, who expects to receive his masters degree in June 1979 under Dr. D. A. Cotter, U. of Windsor, Windsor, Ontario, seeks employment. Interests: Biochemistry and physiology of fungi, medically important fungi.

Michael Dunn is looking for an academic or postdoctoral position. He will be finishing his PhD with Dr. Bandoni at the the University of British Columbia. Research interests include the taxonomy of leaf decay imperfections, especially tropical tuberculariaceous ones, and wood decaying Basidiomycetes.

Fred van der Bogart, who completed his PhD with D. Stuntz in 1975, seeks a teaching or research position. He has teaching competence in introductory biology, botany, Mycology, and does research on the systematics of dark-spored agarics.

David H. Griffin, who expects to complete his PhD Spring, 1980 at the college of Forestry, Syracuse University, seeks a research position in physiological plant pathology.

PUBLICATIONS NEEDED

* Hin Chung Wong

Any publications of the chemistry, biochemistry, and morphogenesis of fungi.

* Robert E. Machol

19th century and earlier works, including fragments, on fungi.

* Barbara Dyko

Walker, J. C. Diseases of Vegetable crops

Chupp, C. & A. F. Sherf. Vegetable diseases and their control.

Sprague, R. 1950. Diseases of cereals and grasses in North America.

* Edith Lea Schnall

Sabouraudia Journals, vol. 1 to vol. 5, part 2.

* Michael Tansey

Cooney & Emerson, 1964. Thermophilic Fungi.

* Guy W. Karr

Publications on members of the Saprolegniaceae

* Donald Pfister

R. W. G. Dennis, *British Ascomycetes*. 1968 edition.

* Edwin Davies

Gertrude S. Burlinghams's monograph of the genus Russula in North America, published in *Flora* 9: 201-236.

H. G. Beardslee's paper in the Mitchell Society journal, 1918, on the North American species.

* Douglas Wright

L. W. Miller & J. S. Boyle: *The Hydnaceae of Iowa*. *Iowa Studies Nat. Hist.* 18(2): (1943) and other papers pertaining to the resupinate Hydnaceae.

* M. R. McGinnis

Lodder, J. & N. J. W. Kreger van Rij (1952). *The Yeasts*. 2nd ed.

* Chris Walker

Kinder, D. A. & M. F. Brown. Electron microscopy of yellow poplar I-IV. *Can. J. Microbiol.* 21: 989-993, 1968-1970, 1930-1939 (1975) and 22: 64-75 (1976)

CHANGES IN AFFILIATION

Walter J. Kaiser was transferred by the USDA from Kenya, East Africa, to the Regional Plant Introduction Station, Washington State University, Pullman.

John Clausz accepted a position in the Biology Department at Carroll College, Waukesha, Illinois.

Barbara Dyko finished a National Science Foundation postdoctoral fellowship at the Commonwealth Mycological Institute, Kew, working with B. C. Sutton, and is now a research associate in Plant Pathology at Michigan State University.

Joseph Ammirati is moving from the Erindale College of the University of Toronto, Ontario, to the Department of Botany, University of Washington, Seattle.

Robert Gessner is moving from the Institute of Marine Sciences, University of North Carolina, to the Department of Biological Sciences, Western Illinois University.

Kerry O'Donnel is now a postdoctoral fellow with David J. McLaughlin, University of Minnesota.

Dennis Gaunt is now a clinical Mycologist with the University Hygienic Laboratory, University of Iowa.

John F. Leslie, having completed a PhD at the University of Wisconsin, working on the genetics and physiology of Schizophyllum, is now working with David Perkins at Stanford University on the genetics of Neurospora.

SEMINARS, AWARDS, HONORS, PROMOTIONS

L. L. Burpee presented a paper on "Differentiation of Rhizoctonia associated with Gramineae" at a symposium on turfgrass diseases in Columbus, Ohio, 16.V.1979.

Bryce Kendrick has been awarded a Guggenheim fellowship for his upcoming sabbatical in Australia, New Zealand, and Polynesia.

Howard Whisler, University of Washington, visited Charles Miller, Ohio University, and presented a departmental seminar.

G. Morgan Jones has been promoted to the rank of Professor of Mycology at Auburn University.

Darryl Grund was promoted from Associate to Full Professor at Acadia University.

Rolf Singer has been appointed Honorary Curator at the Museo Nacional, San José, Costa Rica.

W. L. Steffins and J. P. Jones presented a paper on "Some ultrastructural aspects of ascospore delimitation in Eleutherascus" at the joint symposium of the Texas and Louisiana Society for Electron Microscopy, New Orleans, February, 1979.

George Templeton received the John W. White award for excellence in agricultural research presented annually by the University of Arkansas College of Agriculture, for his work with Colletotrichum gloeosporioides as an agent to control rice field weeds.

Alan Graham and the Department of Biological Sciences, Kent State University, hosted a meeting of a group of 11 persons interested in the study of fossil fungal spores. W. C. Elsik, of Houston, Texas, chaired the group. An illustrated glossary of fossil fungal spores is in preparation.

L. H. Huang gave a talk on "Taxonomy of Actinomycetes" at the Department of Botany, National Taiwan University, Taipei, on 21.V.1979.

Harold Keller has been promoted to Associate Professor, Department of Microbiology and Immunology, Wright State University, Dayton, Ohio.

TRAVELLING MYCOLOGISTS

Dr. George Llano and Dr. Robert L. Gilbertson visited Rolf Singer at the Field Museum of Natural History in May, 1979.

R. T. Hanlin visited the Department of Plant Pathology, University of Arkansas, and presented a seminar on Techniques in Teaching Introductory Mycology.

Dr. Takashi Matsushima, Kobe, Japan, is on a three month visit to the laboratory of G. Morgan-Jones at Auburn University.

Kent Dumont, Dick Homola, and Linda Kohn visited the Farlow Herbarium in the first half of 1979.

Chester Benjamin was a member of the U. S. delegation to the 5th session of the Committee on Agriculture, Food, and Agriculture Organization of the United Nations (FAO), which met in Rome, Italy, April 18-27, 1979. One of the important agenda items was "Review of the International Plant Protection Convention." Some minor revisions were recommended for adoption by the FAO conference in the Fall.

VITAL STATISTICS

Ken and Cindy Conway announce the birth of their second daughter, Deanna Lynne, on January 10, 1979.

Winifred Butterfield married Carroll B. Atkins in Rockville Maryland on March 8, 1978.

OTHER MISCELLANEOUS ITEMS

Michael R. Tansey is collecting material for the second edition of Mycological Teaching Humor.

Jan and Erika Kohlmeyer would like to call attention to "Marine Mycology: The Higher Fungi" to be published by Academic Press this Summer.

Members of the Ohio Mushroom Society are willing to collect fleshy fungi of specific genera in Ohio and Western Pennsylvania, with macroscopic notes, in exchange

for identification aid in these same genera.

The journal *McIlvainea* extends a special call for papers to all persons interested in contributing to American amateur Mycology. Consult recent issues for the range of subjects covered. They are particularly interested in foray reports, biographical material, field observations, Mycological Hobbies, and technical and nontechnical articles dealing with all groups of fungi. Manuscripts should be sent to Richard R. Doyle, Editor-in-Chief, Department of Chemistry, Denison University, Granville, Ohio 42023.

The following two lengthy miscellaneous items were submitted to the MSA newsletter. They are printed in their entirety.

(A). Sectioning small objects with a freezing microtome (A. Y. Rossman)

In the last decade the cellular structure of fungal tissues such as apothecial margins and walls of ascocarps has become an increasingly important taxonomic character. One needs to be able to determine such structure easily and quickly when examining specimens for identification. The freezing microtome has provided the means to do this. I have developed a simple method for embedding and handling small perithecia for sectioning on the freezing microtome which may be of use to others.

Manipulating small objects including thin sections can be done skillfully with insect pins. These are available in various sizes, some very fine and obtainable from any entomology supply store for an extremely low price (35 ¢ per 100 here at Cornell). If the head is removed they can be forced into a wooden matchstick which forms a handle but I find them easy to use without a handle.

Begin by picking the rehydrated specimen off the substrate and placing it on a piece of solid agar. I frequently use areas free of fungi on contaminated plates. Make a small slit in the agar with a needle and insert the specimen in the slit, orienting it so that you will obtain sections through the ostiole or as desired. In sectioning perithecia, only one or two sections will include the ostiole, so orientation becomes crucial. Gently push the specimen into the agar so that it will not change orientation. Cut a small block of agar with the specimen out of the plate. The block of agar with a flat bottom is easy to set on the freezing drop of mucilage in the correct orientation. As the sections are cut, place each one with its surrounding agar directly into a small drop of mounting medium. Sectioned at 15 µm, a perithecium 150 µm diam. will only yield ten sections which fit easily into a drop. When making sections of a type specimen with only a few ascocarps remaining, two good slides can be obtained by placing sections alternately on slides. Thus each slide will have a section through or near the ostiole. At the dissecting microscope inspect the sections in the drop, straightening any crooked or folded ones using an insect pin. Carefully place a cover slip on the drop and hold the slide against the back of a lamp or briefly over a gentle flame. This will melt the agar and thoroughly flatten the cover slip. Placing the sections immediately into the mounting medium eliminates the need of tedious handling of small sections. This method is relatively free of frustration and yields good sections. -Amy Rossman-

Comments from *Mycologia's* Book Review Editor.

Members of the society may wonder why the republication of Enile Boudier's *ICONES MYCOLOGICAE* was given two reviews in *Mycologia*. Both were prepared by me. The first was critical of reproduction, such that it might have discouraged some prospective purchasers. By good fortune the editors in France obtained the skills of M. Jacques Michot, who has taken over the reproduction of the plates. Anyone reading my second review will recognize it as an enthusiastic huzzah!

Ever since I was introduced to mycology at the hands of H. M. Fitzpatrick, I have been aware of the problems of publication and the quality of printing that is necessary to accomplish great plate production. I remain convinced that no single work in mycology is more artistically or more beautifully presented than is the

Boudier ICONES MYCOLOGICAE. (That two of the three volumes of plates are devoted to discomycetes may incline some readers to detect a personal bias on my part. Though I cannot deny that bias, I am not convinced whether or not it was first seeing the Boudier plates that led me to my life studying them, or whether I simply found the discomycetes intriguing on their own.)

With ordinary textbooks and manuals now in the \$60 to \$100 range, the 4-volume set of ICONES MYCOLOGICAE is a bargain. Encourage your librarian to enter a subscription. Future Mycologists will love you for it. -R. P. Korf-

ADDENDA

C. F. Yarwood received an honorary Doctor of Science from the University of British Columbia, June, 1979, and presented two seminars at the same institution.

Harold W. Keller was promoted to Adjunct Associate professor, Wright State University, Dayton, Ohio.

Paul Lemke has been elected to fellowship in the American Academy of Microbiology, and has also been elected president of the Society for Industrial Microbiology.

Walter Sundberg would like to receive specimens of Lepiota sensu lato, and is willing to provide identifications of the same.

A teaching assistantship is available with Walter Sundberg at Southern Illinois University, beginning fall, 1980 - contact W. J. Sundberg.

Richard A. Roeper is studying the Physiology of *Ambrosia* fungi associated with the beetle Xyleborus affinis.

Alwyn Gaetner is studying the lower marine fungi of a Norwegian fjord of high salinity, focussing on taxonomy and ecology. He has cultures of marine Chytrids and Thraustochytrids available for distribution, and would like to receive axenic cultures of lower marine fungi, particularly Thraustochytrids. Dr. Gaetner, Dr. U. Ulken, Dr. K. Schaumann, and Dr. S. Raghu Kumar will attend the International Marine Mycology Symposium, Morehead City, in September, 1979.

MYCOLOGICAL LEVITY DEPARTMENT

(Poem sent to Roland Thaxter, on the occasion of his 70th birthday, by S. F. Blake, Bureau of Plant Industry, Washington, DC, August, 1928)

There was a little beetle once no bigger than a flea:
Like other bugs, he had six legs, and used them constantly.

He travelled up, he travelled down, he travelled in between;
He sampled many things decayed, and many that were green.

He journeyed north, he journeyed south, he followed isotherms;
He gathered many kinds of food, and many kinds of germs.

His very latest day on earth he passed upon a wattle;
A student saw him sitting there and clapped him in a bottle.

He gathered others, some of which a full three inches measure;
But of them all, both great and small, this beetle was the treasure.

(Now you must know, a beetle's wing, just like his leg and tail, is
Created but to be a site for Laboulbeniales).

And when this beetle studied was, upon him there were seen,
Of good new species twenty-five, of genera sixteen.

I saw him lately, where he sits, impaled upon a pin.
He sang a little song to me, and thus it did begin:

"Oh, I've got Labouls on my tarsi, and Labouls on my toes,
On my antennae, in my tubes, and eke upon my nose.

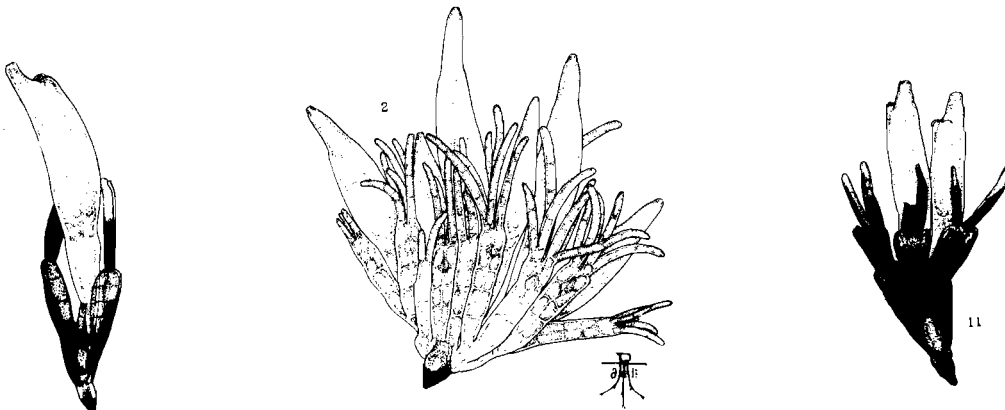
"Upon my tail the master found *Tettigomyces varicus*;
For *T. pterophilus* my wing is still the locus classicus.

"Eleven other *Tettigo*'s from off my abdomen
Were scraped, and sorted into groups, and given names, and then

"The master thought he had enough, and let his needles fall;
But inter nos, he missed the finest Laboulben of all.

"Within my third left spiracle a curious eye would see;
This curious form, which must, I think, form a new family.

"Its foot has toes with well-formed nails, its antherozoids can swim;
And the ascus holds a single spore, with a strongly chitinized rim."



MICHAEL F. ALLEN, PETER D. STAHL*, AND WILLIAM K. SMITH, Department of Botany, University of Wyoming, Laramie, Wyoming 82071 -- Increased transpiration and photosynthetic rates in *Bouteloua gracilis* infected with *Glomus fasciculatus*.

Transpiration (J_w) and photosynthesis ($J-CO_2$) of 3-4 mo-old vesicular-arbuscular mycorrhizal and nonmycorrhizal *Bouteloua gracilis* seedlings were compared. Plants were grown in greenhouse pots containing steam-sterilized prairie soil with sterilized or viable *Glomus fasciculatus* spores. Fifty percent of the root system of each plant inoculated was infected. Soil water potential (Ψ_s), leaf water potential (Ψ_L), leaf resistance (R_L), leaf temp., and CO_2 flux were quantified. J_w and $J-CO_2$ were calculated using electrical circuit analogy equations. J_w doubled with infection as Ψ_s dried from Ψ_w to -6.5 MPa. R_L was 3x greater in saturated soil and increased 3x faster as the soil dried in control than mycorrhizal plants. $J-CO_2$ increased 29% in mycorrhizal over nonmycorrhizal plants at soil saturation. 30% of this increase was due to reduced stomatal resistance to CO_2 uptake and the rest to reductions in mesophyll resistance. These results demonstrate that mycorrhizal infection increases water uptake and CO_2 fixation thus resulting in greater plant productivity.

ALLEN H. ALTMAN* and DAVID H. GRIFFIN. Department of Environmental and Forest Biology, College of Environmental Science and Forestry, Syracuse, N.Y. 13210. Amino acid uptake in *Achlya bisexualis*.

The uptake of six amino acids into germinated spores of *Achlya bisexualis* was studied. The spores were germinated in a defined medium composed of glucose, NH_4 tartrate, thiosulfate and salts. Uptake was measured in this medium. Each amino acid was absorbed in a linear fashion for the length of all experiments. At the end of each experiment more than 90% of the absorbed amino acid remained unchanged. Uptake was inhibited by cyanide, dinitrophenol and azide. The kinetics of uptake follow a Michaelis-Menten relationship with a K_m (μM) and V_{max} (fmole/min/spore) respectively as follows: L-methionine, 0.248 and 1.18; L-arginine, 0.522 and 1.98; L-phenylalanine, 0.641 and 2.88; L-proline, 5.65 and 2.24; glycine, 9.42 and 5.16; L-glutamic acid, 28.9 and 0.892. The low K_m s for L-methionine, L-arginine and L-phenylalanine make them among the highest affinity amino acid transport systems on record. The uptake of each of the six amino acids was inhibited differently by 20 unlabelled amino acids. The kinetics of the inhibition with several reciprocal combinations among the six amino acids showed mixed competitive and non-competitive inhibition.

BACON*, C.W., PORTER, J.K., and ROBBINS, J.D.

Field Crops Laboratory, USDA/SEA, R. B. Russell Agricultural Research Center
Athens, Georgia 30604

Cultural Biotypes of Four *Balansia* Species

Poisoning from ergot alkaloids are normally associated with *Claviceps*; however, we have established that other members of the Clavicipitaceae are capable of producing the ergot alkaloids. Four systemic grass pathogens, *Balansia claviceps*, *B. epichloë*, *B. henningsiana*, and *B. strangulans* were isolated and cultured on defined media. On one medium these fungi produced the ergot alkaloids agroclavine, chanoclavine, elymoclavine, ergonovine, and ergonovinine. Isolates from a given host varied in their ability to produce alkaloids qualitatively and quantitatively. In culture all species had a growth requirement for thiamin and pyridoxine. The addition of nicotinic acid to cultures of *B. epichloë* and *B. henningsiana* caused a tenfold increase in total alkaloid yield but did not increase alkaloid synthesis in other species. Isolates of the four species varied in their pH and temperature optima and in their carbon and nitrogen requirements. In one species, *B. henningsiana*, alkaloid production appears to be host specific. Data suggest that *in vivo* production of alkaloids parallels *in vitro* synthesis.

JOHN G. BAKER*. Analytab Products, Plainview, New York. Medically Important Fungi in the Introductory Mycology Course - Yeasts.

Traditionally introductory mycology courses make little or no mention of medically important fungi in their course outline. Recent improvements in medical techniques have accounted for a decline in the relative occurrence of the virulent epidemic diseases associated with man in past decades and a corresponding increase in the relative importance of opportunistic infections. Yeast infections, comprising the bulk of all fungal infections, may be classified as opportunistic. Classical mycology should reflect this new trend. The environment for communication between classical and medical fields must be improved accordingly. Early exposure to medical mycology would serve to improve knowledge in the area. The influence of medically important yeasts relates directly to the individual. Direct influence of subject matter on the student stimulates interest. Furthermore, yeasts provide excellent laboratory tools for relating taxonomic variation, morphology, physiology, genetics and phylogeny. Inclusion of medically important yeasts in classical morphology should serve to improve the future climate of mycology in both traditional and medical fields.

KAREN K. BAKER and GARY R. HOOPER, Center for Electron Optics, Department of Entomology, Pesticide Research Center, Michigan State University, East Lansing, MI 48824. Ultrastructural development of merosporangia of Piptocephalis indica.

Transmission and scanning electron microscopy were used to study the ultrastructural development of merosporangia of Piptocephalis indica. Standard fixation procedures were used including critical point drying for the scanning samples. Merosporangial branches were initiated from heart-shaped basal spore initials on dichotomously branched sporangioophores. After elongation, spores were cleaved out of the merosporangial protoplast by simultaneous invagination of the plasmalemma. At the completion of cleavage, the plasmalemma fused at the center delimiting more or less equal-sized spores in a single row. At maturity, spores had an electron transparent inner wall layer and an electron dense outer wall layer. The outer wall layer of the end walls was roughened. At spore detachment, the fertile branches had the same roughened appearance as the end walls of the spores and, internally, were highly vacuolated.

EVERETT S. BENEKE, Departments of Botany & Plant Pathology, and Microbiology, Michigan State University, East Lansing, Michigan 48824. Medically Important Fungi in the Introductory Mycology Course - Zygomycetes. A number of the genera in the Zygomycetes have become increasingly important to include in the study in an introductory course in mycology. Various species in the order Mucorales may be the causative agents of mucormycosis in debilitated patients. Species of Absidia, Mucor, and Rhizopus often develop massive invasion of the blood vessels, central nervous systems or nasal sinuses, and are usually rapidly fatal in these patients. Several species in the order Mucorales may be the cause of abortion in cattle and swine. Still others are valuable in the fermentation of steroid products. There are several species in the order Entomophthorales that produce granulomas in the subcutaneous tissue or nasal polyps in man and in the horse. Some of the Zygomycetes are known to produce toxins in grains and foods when growing in conditions suitable for degradation.

GERALD L. BENNY,* DON A. SAMUELSON, and JAMES W. KIMBROUGH. Dept. of Botany, Univ. of Florida, Gainesville, FL 32611. Light and Electron Microscopic Studies of Ascocarp and Wall Ontogeny of an Undescribed Species of Chaetomidium with a Cephalothecoid Peridium.

Many species of Plectomycetes are known that produce a cephalothecoid peridium but little is known about the ontogeny of this wall type. The isolation of an undescribed Chaetomidium with a cephalothecoid wall, that sporulated well in culture, presented an opportunity to study the ontogeny of this peridial type. The fungus was

grown on corn meal agar (Difco) at 25 C. Whole mounts of the fungus were made in either KOH-Phloxine or Lactofuchsin. Sections of ascocarps produced in culture were made with an International CTI Cryostat. Observations were made with either phase contrast or brightfield microscopy. Dried cultured material was shadowed with gold-palladium in a Hummer I sputter coater and viewed with a Omniscan Scanning Electron Microscope. The ascocarp appendages begin to appear very early in the ontogeny of the ascocarp. The asci are broadly clavate, short-stipitate, and eight-spored. The ascospores are smooth-walled, light brown, and produce a single germ pore. The cephalothecoid plates become differentiated late in ascocarp ontogeny, and are generally produced around the base of a single ascocarp appendage. Ascocarp and peridial ontogeny are discussed and illustrated in detail.

Donald A. Betterley* and O'Neil Ray Collins. Department of Botany, University of California, Berkeley, CA 94720. - Enzyme polymorphism and reproductive systems in the slime mold Didymium (Myxomycetes).

Enzyme polymorphism is used as a measure of the effects of different reproductive systems on the amount of genetic variation within and between isolates of the myxomycete Didymium iridis. Isolates are classified as either heterothallic or non-heterothallic, and plasmodial extracts are used for isozyme analysis by polyacrylamide step-gradient gel electrophoresis. A variety of mechanical extraction methods were used in combination with a range of protease inhibitors in the extraction buffer. Neither food source nor the extraction methods have altered the isozyme patterns for the enzymes GOT, G6PDH, MDH, LAP, acid phosphatase, phenoloxidase or esterases. Plasmodial aging up to 150 days also has no effect on isozyme patterns. There is little genetic divergence among the 7 known interfertile heterothallic isolates, whereas there is greater divergence among the 17 non-heterothallic isolates. The non-heterothallic forms are not known to interbreed. Results from the non-heterothallic isolates suggest that in nature there are many genetically distinct and reproductively isolated populations. Morphological characters are essentially identical for all of the above isolates, and plasmodial compatibility tests have shown that none of the isolates are somatically compatible. Isolates of Didymium megalosporum, nigripes, difforme, saturnus and squamulosum are included in the isozyme studies as a measure of interspecific variation.

MEREDITH BLACKWELL. Department of Biology, Hope College, Holland MI 49423. Distribution, Host Specificity, and Variation in Arthrorhynchus nycteribiae and A. eucampsipodae (Laboulbeniomyces).

The genus Arthrorhynchus is known only from wingless Old World dipteran parasites of bats (Nycteribiidae). Although there are a number of records of these fungi from the literature, this report of

twenty-one additional collections of *A. nycteribiae* and fourteen of *A. eucampsipodae* substantially increases the geographical and host ranges of both species. While *Arthrorhynchus* is restricted to this family of bat flies, there does not seem to be strict specificity for certain fly species. Bat habit does appear to be important, and infected flies were found on gregarious, cave roosting species. Marked intraspecific variation is consistent with small population sizes and extreme isolation observed in these fungi.

BEVERLY BONNELL* and ESTELLE LEVETIN, Faculty of Natural Sciences, University of Tulsa, Tulsa, OK 74104. The effect of light and medium on growth and sporulation of two strains of *Epicoccum nigrum* Link.

Two strains of *Epicoccum nigrum* Link were cultured on potato dextrose agar and on nine test media, each with three carbon concentrations, in three light regimes. Colony diameter, total number of spores, and spore density were computed. On potato dextrose agar, growth and sporulation of both strains was enhanced by light. The sporulating strain averaged thirteen times more spores and 25% greater size than the mycelial strain. On all the test media, growth and sporulation averages for the two strains were much closer, with slightly larger size in darkness. Starch and sodium nitrate were best nutrient sources for the mycelial strain, glucose and asparagine for the sporulating strain. Sodium acetate and ammonium sulfate were poor nutrient sources for growth and sporulation in both strains. Previous assertions that light is required for sporulation were contradicted, but the reported variable nature of the species was illustrated.

FRANK A. CANTONE*, E. A. VALENTINE, AND S. E. GOCHENAUR, Biology Department, Adelphi University, Garden City, New York

Microfungi from the carposphere of *Laccaria trullisata* and their relationship to the bare soil community

The arenicolous basidiomycete, *Laccaria trullisata*, grows partially submerged in sand along the shore of Hempstead Lake, L.I., N.Y. and functions as a concentrated carbon-energy source for other heterotrophs in this pioneer ecosystem. This study examines its effect on the soil microfungal community. Populations isolated using dilution and soil plate methods from carpospheres (sand adhering to the basidiocarp) and bare soil collected 50 and 100mm from the mushrooms were compared. Yeasts, sterile taxa, pycnidial forms and species diversity increased in the carpospheres; the penicillia decreased. The latter accounted for 80% of the bare soil isolates but less than 60% of those obtained from sand adjacent to the mushrooms. The responses of the individual penicillia differed. *P. terlikowskii* and *P. cyclopium* were more abundant in the bare soil; *P. funiculosum* and *P. duclauxii* in the carposphere; while the distribution of *P. janthinellum*, the leading dominant, did not change. The results of *in vitro* studies

examining the role of nutrition, environment and antagonistic compounds in generating the carposphere effect will be presented.

COLE, GARRY T. Department of Botany, University of Texas, Austin, Tex. 78712. Cultural studies of conidial fungi pathogenic to man.

The conidial fungi include those imperfect forms which produce conidia from reproductive structures borne on loosely arranged or aggregated hyphae (Form-Class: Hyphomycetes) or reproductive cells enclosed in sporocarps (Form-Class: Coelomycetes). In the latter group, only *Phoma* has been associated with human mycoses (e.g. agent of phaeomycotic cyst). The Hyphomycetes, however, include a plethora of pathogenic and/or opportunistic fungi the identification of which frequently depends on examination of axenic cultures. Certain morphological and developmental criteria are used to recognize and classify these fungi in pure culture. Representative species associated with the major clinical types of mycoses (superficial, cutaneous, subcutaneous, and systemic) are illustrated and discussed.

O'Neil Ray Collins. Department of Botany, University of California, Berkeley, CA 94720. - Life cycles and evolutionary relationships in the Myxomycetes.

Insofar as we know, all heterothallic myxomycete entities possess a one-locus, multiple allelic system of mating, suggesting a monophyletic origin before divergence of extant taxa. Heterothallism may be most prevalent among members of the Physarales, although non-heterothallic forms are also common in the order. For these and other reasons, I tentatively propose that today's Myxomycetes might share a common physaraceous ancestor, perhaps one similar to *Physarum polycephalum*. (Organisms such as *Echinostelium minutum* are viewed as derived, reduced and specialized.) Further, it is presumed that the amoebflagellate stage evolved first, that plasmodial development followed the introduction of syngamy into the life cycle and that sporophore production and meiosis evolved last. Therefore, research which includes studies on the amoebflagellate and plasmodial stages may reveal more about relationships between major taxa than investigations on the sporophore stage alone. Fundamental to this kind of research is an understanding of the taxonomic distribution of heterothallism and non-heterothallism in the class. Correlations between the reproductive cycles and other traits probably exist, and these should be useful in delineating evolutionary relationships both at the species and higher taxonomic levels. These and other phylogenetic views will be discussed and supporting information will be presented.

J. L. CRANE, Illinois Natural History Survey, Urbana, IL 61801, C.A. SHEARER, Botany Dept., University of Illinois, Urbana, IL 61801. The Comparative Morphology of Two Cleistothecial Ascomycetes with Papilionaceous Ascospores.

Argyryna polyhedron (Schw.) Morgan is characterized by globose, astomatous, dark brown, sessile or stalked ascocarps with characteristic hexagonal peridial plates. The asci are clavate and develop on slender stalks which in mass form a dense basal hymenium. Mature ascospores are papilionaceous, 2-celled, light brown with a dark band at the septum and measure $4 \times 7 \mu\text{m}$. Ascocarp dehiscence in A. polyhedron is by separation of the apical peridial plates exposing the spore mass in a cup-shaped basal cavity. A deceptively similar ascomycete, C.S. 470-1, representing a new genus and species was isolated from submerged, decayed wood in freshwater. The ascocarps are membranous, globose, pseudo-parenchymatous, dark brown and lack the hexagonal plates. The asci are oblong to clavate and formed in a basal fascicle. The ascospores are papilionaceous, 2-celled, light brown and measure $24 \times 30 \mu\text{m}$.

FREDERIC H. ERBISCH, Department of Biological Sciences and the Keweenaw Research Center, Michigan Technological University, Houghton, Michigan. - Armillariella mellea, its occurrence and distribution in upper Michigan mines. Two years ago a large mass of fungal rhizomorphs was found in an abandoned copper mine. The rhizomorphic fungus was, after isolation, identified as Armillariella mellea. Subsequently, in further explorations, more rhizomorphs of this fungus were found in other copper and iron mines. Although suitable substrates were found throughout the mines, the fungus displayed a limited distribution. The distribution does not appear to be limited by the type of substrate, but rather by moisture availability. Sites favorable for A. mellea, usually had no or few other fungi associated with the rhizomorphs. In sites not favorable for A. mellea other fungi were abundantly evident.

PAUL H. DUNN. Pacific Southwest Forest and Range Experiment Station, U.S. Forest Service 110 North Wabash Avenue, Glendora, CA 91740. - The Relationship of Chaparral Stand Age to Litter Decomposition rates.

An IRGA study of several ages (since last fire) of chaparral in lab and field showed differences in decomposition rates. The older the stands, the lower the decomposition rates of the litter. These differences could not be attributed to changes in C:N ratio. The changes in decomposition rates are a major factor to be considered in determination of the age of burning in chaparral prescribed burning programs.

STANLEY L. FLEGLER*, GARY R. HOOPER
Pesticide Research Center
Michigan State University
East Lansing, MI 48824
Ultrastructure and Development of Nidularia pulvinata

No ultrastructural developmental studies have been done on the primitive members of the Nidulariaceae. This study was initiated to learn more about these fungi. Nidularia pulvinata ba-

sidiocarps were grown in the laboratory on a special nutrient supplemented straw medium. Basidiocarps were freeze-sectioned in a cryostat, fixed, dehydrated, critical point dried, and observed in the SEM. Young basidiocarps were clearly differentiated into an upper and lower portion. Peridiole formation began in the lower portion. The wall of the young basidiocarps was composed of dense spinose skeletal hyphae and later differentiated into three layers. The peridioles were formed from round aggregations of hyphae. The peridioles gradually became oval shaped with a central cavity in which basidiospores were produced. Four basidiospores on long sterigmata were formed on each basidium. The wall of each peridiole was composed of two layers. The pattern of development as revealed in our study differs from that in previously reported studies.

R.A. FROMTLING*, R. BLACKSTOCK, and G.S. BULMER. Departments of Microbiology and Immunology, and Pediatrics, University of Oklahoma Medical Center, P.O.B. 26901, Oklahoma City, Okla. 73190. Cryptococcus neoformans: Immunization and Passive Transfer of Immunity in Mice.

We have developed a murine model for studying the cell-mediated immune response in experimental cryptococcosis. Groups of mice (C57/B16J) were given 2, 4, 6, or 8 weekly intraperitoneal inoculations of a live, avirulent, pseudohyphal form of Cryptococcus neoformans (NU-2-P). Spleen cells from immunized animals were tested for cryptococcal antigen sensitivity via an *in vitro* lymphocyte transformation (LT) assay. Spleen cells from the 6 and 8 week treated groups showed significantly increased stimulation to several killed strains of C. neoformans in comparison to the saline control, 2, and 4 week treated mice. Significant survival was seen in 6 and 8 week treated mice following lethal intravenous (IV) challenge with C. neoformans. Spleen cells and serum from another group of 8 week treated (immunized) and non-treated (control) mice were transferred to normal animals by intraperitoneal inoculations. Following lethal IV challenge, immune spleen cell treated mice showed extended survival in comparison to immune serum, normal serum, and normal spleen cells treated groups. A positive correlation ($r = .90$; $p < .05$) between increased LT response and survival of IV challenge was established, as was passive transfer of immunity in experimental murine cryptococcosis.

R.A. FROMTLING*, B.E. ROBINSON, J. JENSEN, and G.S. BULMER. Department of Microbiology and Immunology, University of Oklahoma Medical Center, P.O.B. 26901, Oklahoma City, Okla. 73190. Pathogenesis of Beauveria bassiana in Captive American Alligators.

Three captive American alligators (Alligator mississippiensis) at the Oklahoma City Zoo developed severe pulmonary disease and succumbed. Two died immediately following an extended hibernation and one survived the summer, but expired in the early fall. Post-mortem examination revealed extensive pulmonary involvement with numerous white fluffy lesions surrounded by necrosed consolidated tissue. Histopathologic examination showed

septate, branching hyphae which had blocked the bronchial and alveolar spaces, and had invaded adjacent tissues. Dissemination to the liver and spleen had occurred in 2 of the 3 animals, with the remaining case being only pulmonary. Fungal colonies measuring 3 square cm were found on the pleura of one animal. These colonies had sporulated *in vivo* as detected by examination of histopathologic sections. The entomopathogenic fungus, *Beauveria bassiana*, was cultured from the lung tissue of each alligator. When grown on Sabouraud's dextrose agar (SDA) at 28 C, the organism produced a red-brown diffusible pigment and formed a colony 2 cm in diameter after 20 days. The fungus failed to grow on SDA or brain heart infusion agar at 37 C. *B. bassiana* is a well recognized agent of fungal disease of insects. It has been reported only rarely as an agent of vertebrate mycosis.

M.S. FULLER* and J.W. TAYLOR Department of Botany, University of Georgia, Athens, GA 30602. Operculum and suboperculum formation in *Chytridium confervae*.

The usefulness of operculum as a taxonomic characteristic is currently in question (Barr, 1975. Can. J. Bot. 53:164). We are using light and electron microscopy to study zoosporogenesis, especially the formation of the operculum and suboperculum (hyaline globule of Barr, *ibid.*), in *Chytridium confervae* (Wille) Minden, a monocentric, eucarpic chytrid. During zoosporogenesis material is deposited between the plasma membrane and the sporangium wall adjacent to the future site of the operculum. This suboperculum is composed of a thin layer adjacent to the cell wall and a larger, hemispherical mass of loose fibrous material that abuts the plasma membrane. The suboperculum is periodic acid Schiff's positive. Lead and uranyl acetate stains leave the suboperculum electron transparent. Prestaining with barium permanganate (Hoch, 1977. Mycologia 69:1209) makes both the thin layer and a reticulum of fibers in the hemispherical mass electron opaque. Periodic acid followed by either phosphotungstic and chromic acids (Lembi, et al., 1971. Planta 99:37) or silver proteinate (Thiery, 1967. J. Microscopie 6: 987) leaves the thin layer electron transparent and makes the fibers of the hemispherical mass electron opaque. Vesicles that arise from Golgi membranes appear to be the source of the suboperculum. The suboperculum is apparent 3h after the induction of zoosporogenesis by starvation and is complete just prior to cleavage of zoospores, 4.5h after induction. At 6.5h, the suboperculum disperses and the discharging zoospores are constrained by a vesicle.

P. K. GAUR* and R. W. LICHTWARDT, Department of Botany, University of Kansas, Lawrence KS 66045.

Isozyme variation in natural populations of *Histoplasma capsulatum*.

Phenotypic variation was measured in 369 isolates of *H. capsulatum* obtained from 3 starling roosts in Missouri and Kentucky and 1 seagull nesting site in Michigan. Identification of the isolates was based on micromorphology, ability to mate with tester strains of *Ajellomyces capsulatus*, and convertibility to the yeast phase *in vitro*.

The 8 enzyme systems extracted by sonication and electrophoresed on starch gels were: leucine aminopeptidase (LAP), acid phosphatase (ACP), alkaline phosphatase (ALKP), α -esterase (EST), phosphoglucomutase (PGM), phosphoglucoisomerase (PGI), glucose dehydrogenase (GDH), and alanine dehydrogenase (ADH).

Populations differed geographically, and considerable polymorphism was detected in some enzyme systems within populations and even among isolates from the same soil sample. LAP, PGM and PGI were most useful to detect differences. Five phenotypes (isozyme band patterns) were found only in isolates of the (+) mating type, and 2 were found only in (-) mating types.

Isozyme patterns were consistent in replicate runs of the same isolates. They offer a sensitive method to detect phenotypic variation in populations not measurable by other means.

P. K. GAUR* and R. W. LICHTWARDT, Department of Botany, University of Kansas, Lawrence KS 66045.

Mating types and convertibility among soil isolates of *Histoplasma capsulatum*.

The overall ratio of mating types in 335 isolates of *H. capsulatum* (*Ajellomyces capsulatus*) from 3 starling roosts and 1 seagull nesting site was 54% (-) to 46% (+). However, the ratios varied considerably among the 4 sites and among samples collected within the same site. A starling roost in Warsaw, MO (2 soil samples, 42 isolates) produced 100% (-) and 0% (+) mating types; in Springfield, MO (6 samples, 196 isolates) 42% (-) and 58% (+); and in Hopkinsville, KY (2 samples, 53 isolates) 91% (-) and 9% (+). The seagull site near Rogers City, MI (1 sample, 44 isolates) produced 16% (-) and 84% (+) mating types.

Only 13% of the 335 isolates [6% (-) and 7% (+)] converted well from the filamentous to the yeast phase on enriched medium at 37°C. The other isolates remained predominantly filamentous, but colonies were waxy with a few yeast cells. This percentage of convertibility and the frequencies of mating types are compared with a study of soil and clinical isolates by Kwon-Chung, Weeks and Larsh (1974).

LARRY P. GAURILOFF* AND RONA DELAY. Department of Botany, University of Georgia, Athens, GA 30602. Comparative Ultrastructure and Biochemistry of the Harpochytriales - Taxonomic Implications.

Recent investigations have indicated that the internal morphology of the posteriorly uniflagellate zoospore may be instrumental in demonstrating relationships among chytridiomycetous fungi. Inherent in many of these studies is the apparent confusion caused by the Harpochytriales. In order to clarify this confusion, the zoospores of *Oedogoniomyces*, *Harpochytrium* and the monoblepharidalean fungus *Monoblepharella* were re-examined ultrastructurally using serial sections. The arrangement and structure of most organelles were similar in the zoospores of *Oedogoniomyces* and

Monoblepharella. The only apparent differences between the zoospores of these two organisms were the location of the mitochondria and the structure of their rumposomes. The arrangement of organelles in the zoospores of Harpochytrium was different from the zoospores of Oedogoniomyces and Monoblepharella, although some subtle similarities were noted. The most apparent difference was the presence of a unique microbody-lipid globule complex in the Harpochytrium zoospore that was not present in the zoospores of Oedogoniomyces and Monoblepharella. This data suggested that Oedogoniomyces was more closely related to Monoblepharella than to Harpochytrium. The relative molecular weights of ribosomal RNA supported this contention. The relative molecular weights of the 25s RNA of Oedogoniomyces and Monoblepharella were the same. The relative molecular weight of the Harpochytrium 25s RNA was similar to chytrid rRNA.

ROBERT L. GILBERTSON, Department of Plant Pathology, University of Arizona, Tucson, AZ 85721.
The wood-rotting fungi of North America.
Presidential address.

This paper deals with the basidiomycetes that decay wood in North America. There are an estimated 1650-1700 species distributed in a number of families of Tremellales, Aphyllophorales, and Agaricales. The majority of these are in the Polyporaceae, Corticiaceae, Tricholomataceae, and Cortinariaceae. In general, the species of North American wood-rotting fungi are widely distributed throughout the northern hemisphere. For example, 80-90% of Scandinavian Corticiaceae are also known in North America. Most of the North American wood-rotting fungi are "white-rotters" with enzyme systems that break down cellulose and lignin. A small percentage (about 10%) are brown rotters and do not utilize lignin. Most of these brown rot fungi are in the Polyporaceae with a few additional species scattered through the other families. Almost all brown rot fungi that have been tested have a bipolar mating system and occur primarily on conifers. I interpret them as recently evolved organisms adapted to the low temperatures and drier conditions of coniferous forest ecosystems.

Among the interesting relationships of wood-rotting fungi are those with insects. Symbiotic relationships exist between some North American wood-rotting fungi and wood wasps. Some brown rot fungi, especially Gloeophyllum trabeum, produce termite attractants and some white rots repel termites. Many insects inhabit basidiocarps of wood-rotting fungi, and some, like the Ciid beetles, display interesting degrees of host specificity apparently related to hyphal structure.

Some wood-rotting fungi have edible basidiocarps and a few including Armillariella mellea and Sparassis radicata are among the choicest of edible fungi. Others, however, are highly toxic and are known to contain toxins of the cyclopeptide, coprine, muscarine, or psilocybin-psilocin types as well as gastrointestinal irritants.

In many parts of the world wood is a primary energy source and supplies have been seriously depleted. We are fortunate to have large areas in North America with large volumes of non-commercial wood residues that are largely unused for fuel. As a result

we can still develop a rather complete knowledge of the native wood-rotting fungi. The situation will change in the future and wood as an energy source will become more important as other traditional sources including oil and coal become less available and more expensive. As our large volumes of non-commercial wood residues gradually disappear, the wood-rotting fungus biota will also decrease and eventually many species may be lost from the North American flora. Therefore it is imperative that a vigorous attempt be made to provide a complete treatment of North American wood-rotting fungi now before drastic ecological changes occur as a result of utilization of wood as an energy source.

J. GINNS* and S. SUNHEDE
Biosystematics Res. Inst., Canada Agric.,
Ottawa and Univ. Göteborg, Sweden
Teratological Galls on the Mushroom Collybia dryophila.

Collybia dryophila (Fr.) Kummer has for many years been known to bear galls. The gall formation has been explained as a reaction to excessively wet growing conditions and overmaturity of the mushroom. Our examination of specimens from Canada, Denmark, Sweden and the U.S.A. led to the conclusion that there are three fungi (all species of Christiansenia Hauerslev) parasitizing the mushroom and causing the galls to develop.

S. E. Gochenaur, Biology Department,
Adelphi University, Garden City, N.Y. 11530

Hydrolase patterns among the members of a soil microfungal community

Over 200 isolates representing 53 species were tested for the ability to produce enzymes that hydrolyze the following substrates: gelatin, casein, albumen, triglyceride, diphosphophenolphthalin, nucleic acids, starches, pectin, xylan, arbutin, acid-swollen cellulose and colloidal chitin. These fungi, members of a community described earlier (MYCOLOGIA 70: 975, 1978), were isolated from the A horizon of an oak-birch forest. Strains were tested in duplicate on a solid buffered basal medium with an agar over-layer containing 2% of a substrate. Evaluation of enzyme production by a change in opacity or color of the medium around a colony suggest for the penicillia that a positive correlation exists between the number of hydrolases secreted and importance in the community. Over 75% of the isolates decomposed gelatin, the triglyceride and amylose but few attacked albumen, RNA, pectin at pH 7.2 and arbutin. It is concluded that the microfungal community possesses the potential to degrade a wide variety of organic compounds but its members show more limited capacity and individuality in the patterns of hydrolases produced.

Donna Goldstein* and R. L. Gilbertson, Dept. of Plant Pathology, Univ. of Arizona, Tucson, AZ. 85721. Cultural studies of the wood-rotting fungus Inonotus arizonicus.

Inonotus arizonicus (Aphyllphorales; Hymenochaetaceae) causes a heartrot of Arizona and California sycamores and is known only from the Southwest in those hosts. Like all members of the Hymenochaetaceae, *I. arizonicus* lacks clamp connections. This presents difficult problems in distinguishing homokaryotic and heterokaryotic isolates. *I. arizonicus* fruits readily in culture within 4 weeks on malt extract agar medium and may sporulate heavily for a week or more. Basidiospores germinate readily and give rise to single spore isolates that also fruit and sporulate readily. The basidiocarps from these single spore isolates are morphologically identical macroscopically and microscopically to those that develop from presumptive heterokaryons derived from basidiocarp tissue isolates and mass basidiospore cultures. Preliminary cytological investigations of nuclear state have produced no evidence of differences between single basidiospore isolates and tissue isolates. These observations strongly suggest that *I. arizonicus* is homothallic. Presumptive heterokaryotic tissue isolates from different locations produce a dark interaction zone when grown on the same plate. Homozygous matings do not form an interaction zone. Agar block decay tests show that single spore isolates show no significant differences in decay capacity from tissue isolates. *I. arizonicus* readily decays wood of several associated species (Arizona ash, Arizona black walnut, Southwestern chokecherry, and netleaf oak) although it does not occur on these hosts in nature.

GOTTWALD, T. R.* and H. R. CAMERON. Department of Botany and Plant Pathology, Oregon State University, Corvallis, OR., 97331--Studies in the Morphology and Life History of Anisogramma anomala.

A morphological and cytological study of the life history of Anisogramma anomala (Pk.) E. Muller, which causes a canker disease of commercial hazelnut, Corylus avellanae L. was undertaken. Mycelium colonizes the phloem forming a pressure cushion of pseudoparenchyma which eventually develops into a stroma. Perithecia form from primordia which are scattered throughout the hypostroma. The primordia consist of an 'outer coil' of filamentous hyphal cells terminated by a non-functional trichogyne, surrounding an 'inner coil' of more isodiametric ascogonial cells. The centrum arises from this inner coil as a loose clump of ascogenous cells which lack croziers. Asci arise from mononucleate ascogenous hyphae. The asci go through a typical meiosis followed by two successive mitotic divisions giving rise to eight unequally 2-celled, hyaline ascospores. Spore dissemination occurs in response to hydration of both the ascocarp and individual asci. The pathology and histology of the host's response to infection are discussed.

M. GUNASEKARAN*, DIVISION OF INFECTIOUS DISEASES, ST. JUDE CHILDREN'S RESEARCH HOSPITAL, MEMPHIS, TENNESSEE 38101. Production of Tryptophol and Indole Lactic Acid by Cryptococcus neoformans

Cryptococcus neoformans isolated from a leukemia patient was grown in 10 ml of synthetic

medium (containing 2% glucose, 0.1% L-tryptophan as carbon and nitrogen source along with other essential salts) as an aerobic culture at 37°C. At various time intervals, samples were removed to measure the growth of the organism and to identify products released into the culture medium. Tryptophol (TOL) and indole lactic acid (ILA) were the two main products in diethyl ether extracts of the medium as determined by gas liquid chromatography. The physical and chemical properties of these compounds were confirmed by comparison with known standards. Similar findings were obtained with C. neoformans isolated from different sources. On the basis of these results TOL and ILA from C. neoformans may serve as markers for identification of the organism by gas chromatographic procedure.

I. BRENT HEATH. Biology Department, York University, Toronto, Canada.

Fungal Mitoses, the Significance of Variations on a Theme

Numerous features of mitosis in fungi have been unclear and controversial for many years. Recent research has taken the questions to a higher resolution but it is still clear that fungal nuclei show considerable heterogeneity of behaviour relative to higher plants and animals. These variations may help explain the process of mitosis itself and serve as indicators of evolutionary trends among the fungi and their allies.

The diversity of spindle structure and chromatin behaviour, details of which will be discussed, help differentiate between possibly universal hypotheses for mitotic force generation or, alternatively, provide evidence for the evolution of multiple mitotic mechanisms. Either way, the availability of large, synchronous, populations of small, easily analyzed, mitotic nuclei in organisms amenable to genetic analysis is a feature of many fungi. This feature makes the fungi excellent organisms in which to investigate the mechanisms of mitosis.

In the context of evolution and phylogeny, recent analysis shows that the Mucorales may have the most primitive type of eukaryotic mitosis known. Above this level a consideration of the details of mitotic systems produces some interesting and unexpected groupings of the fungi and other organisms, details of which will be justified.

I. BRENT HEATH*, MARY-LOU ASHTON and MICHELE C. HEATH. Biology Department, York University, 4700 Keele Street, Downsview, Ontario M3J 1P3, Canada and Botany Department, University of Toronto, Toronto, Ontario M5S 1A1, Canada.

Mitosis in three species of Taphrina

The genus Taphrina has been postulated as one similar to a hypothetical ascomycetous ancestor of the basidiomycetes. Among morphologically simple organisms there is a paucity of characters on which to base phylogenetic hypotheses, thus additional characteristics such as details of the

mitotic process may help clarify relationships. Mitosis in agar grown, yeast phase cultures of *T. deformans*, *T. polystichi* and *T. californica* was studied from electron microscopy of serial sections. In both *T. deformans* and *T. polystichi* the details of mitosis are comparable to those of other ascomycetes with disc like NAOs which sink into close fitting pores in the nuclear envelope at mitosis. The nuclear envelope remains intact and mitosis occurs in the mother cell. In contrast the mitotic events occurring in *T. californica*, which parasitizes the fern *Dryopteris arguta*, are very similar in every respect to those of the basidiomycetous yeast *Rhodosporidium* (McCully & Robinow, 1972, J. Cell Sci. 10, 857). Alternative explanations for these observations will be examined.

A.D. Hewings* and C.A. Shearer. Dept. of Botany, Univ. of Ill., Urbana, IL 61801. Seasonal Changes in Aquatic Hyphomycete Community Structure.

A 13 month quantitative distribution study of an aquatic hyphomycete community was made by randomly collecting 100 leaves at monthly intervals at a single site on the Sangamon River, Ill. Ten discs were cut from each leaf, incubated at stream temperature for 3-5 days, fixed, stained and examined for the presence of conidia. Three indices of community structure were determined: (1) diversity (D), (2) Shannon-Weaver diversity (H'), (3) Gini coefficient of concentration. Although the relative importance values of different species changed with respect to one another, diversities (H') were remarkably similar throughout the fall, winter and spring. The major change in species composition, D and H', occurred in the summer months. Seasonal changes in species distribution were analyzed using regression analysis of the form: $Y(i,t) = a + bt$. Two distinct distribution patterns emerged. One is a unimodal, persistent pattern where a species is present throughout the sampling period and exhibits a definite maximum at a particular time of the year. *Tetracladium marchalianum*, *Filosporella annelidica* and *Lemonnieria aquatica* were of this type with the first two species reaching a maximum in the fall and the last species in the winter. The second type of species distribution is unimodal and seasonal in which a species is present from one to two months of the year and absent for the remainder of the sampling period. In late spring, *Lunulospora curvula* and *Triscelophorus monosporus* appeared, reached a maximum in late summer and disappeared abruptly in late fall. Similarly, *Dimorphospora foliicola* was observed only in mid-winter but never in summer. The other patterns were more subtle and difficult to generalize.

TERRY W. HILL* and J. T. MULLINS
Dept. of Botany, Univ. of Florida
Gainesville, Florida 32611

Isolation of cellulase-containing vesicles from *Achlya ambisexualis*

Cellulase (EC 3.2.1.4) has been implicated in cell wall morphogenesis in *Achlya*, and may be delivered to the wall in vesicles. *Achlya* cellulase is an integral membrane protein which can be released by

1 % triton X-100, but not by salts or freezing. In this paper, the possibility is explored that cellulase may be present in apical vesicles of vegetative hyphae. By electron microscopical cytochemistry, apical vesicles were found to exist in two size classes with diameters of about 80 nm or 150 nm; and the larger class is characterized by IDPase activity and a fibrous carbohydrate matrix. Mycelial homogenates were fractionated by differential and sucrose density gradient centrifugations, and IDPase-rich and carbohydrate-rich membranes equilibrated at about 1.19 g/cm³. This was also the point at which maximum membrane-bound cellulase and UDPG transferase activities equilibrated. It is concluded that *Achlya* apical vesicles contain cellulase, UDPG transferase, and carbohydrate; and this supports theories which postulate that apical growth results from a dynamic equilibrium between wall synthesis and wall lysis at the apex.

RICHARD L. HOMOLA* Department of Botany and Plant Pathology, University of Maine, Orono, Maine. 04469. Some morphological variations in the germ pores of basidiospores in the Agaricales.

The basidiospores of some members of the Coprinaceae, Bolbitaceae, Strophariaceae, Lepiotaceae and Boletaceae were examined using the scanning electron microscope (SEM). These families have some species with basidiospores with a germination pore at the apical end. The pores are not as the term would imply an opening, but a depression or a thinning in the spore wall. The SEM has revealed external variation in these pores. The most common and least modified germ pore is a simple depression centrally located at a truncate apical end as found in the Coprinaceae and Bolbitaceae. In the Lepiotaceae the pores are simple, but the apical end is less truncate and the pore smaller. In *Conocybe* sp. more than one eccentrically located germ pore was observed. Of those species examined, *Psathyrella foenicis* had the only roughened basidiospore with a simple pore. Most basidiospores have a single pore. Most basidiospores with simple germ pores had smooth unornamented surfaces. *Psathyrella velutina* has a basidiospore with a different germ pore morphology. The verrucose basidiospore has a tapering almost snout-like unornamented apical end with a raised ring surrounding the germ pore. *Boletellus betula* has a pitted basidiospore with a truncate apex. The centrally located germ pore has a raised margin with internally longitudinal striae. The germ pore is cushion-like and depressed. Germ pores vary in their morphology and thus may be important taxonomically to show phylogenetic relationships.

RICHARD A. HUMBER, Boyce Thompson Institute, Tower Road, Ithaca, New York 14853

Nuclear Cytology as an Indicator of Generic Classification in the Entomophthorales

Controversies over the last 15 years about the correct generic limits within the zygomycete order Entomophthorales have not been resolved using conventional taxonomic techniques. Examinations of representative fungi from all genera of the order indicate that at least three morphologically distinct types of nuclei exist, and that each type has its own characteristic mitotic pattern. These nuclear characteristics are more or less readily detectable with phase-contrast microscopy of live

or preserved material. The three nuclear types belong to biologically distinct groups which very probably deserve recognition as separate families (Entomophthoraceae, Basidiobolaceae, and the Ancylistaceae).

Nuclear types confirm that the genus Culicicola Batko should be rejected as a mixture of species belonging to Entomophthora sensu stricto and to Conidiobolus, and that Entomophaga Batko (as originally described) also contains some species of Conidiobolus. Following the recognition and transfer of the entomogenous species of Conidiobolus, the rejection of Culicicola, and the emendations of Entomophaga and Entomophthora s. str., an orderly and sensible re-arrangement of the species of this order along lines similar to those proposed earlier by Batko will become practical.

Jenkins, David T. and Jones, Daniel D., Biology Department, University of Alabama in Birmingham, Birmingham, AL 35294; Hatfield, George M., College of Pharmacy, University of Michigan, Ann Arbor, MI 48109; Cibula, William G., National Space Technology Laboratories, Bay St. Louis, MS 39520

Laboratory cultivation of Amanita specimens. The general morphology of Amanita specimens is familiar to many mycologists. The nature of its nutritional requirements, however, has been given little attention. Its specimens are known to be mycorrhizal, thereby implying some of its nutritional needs. It is the purpose of this study to elucidate the nutritional requirements for cultivation of Amanita specimens in the laboratory. Many Amanita specimens will grow vegetatively in laboratory cultures, but none have been induced to fruit. This lack of success may logically be attributed to the inability to provide growth conditions associated with its natural mycorrhizal state. In phase one of this study media combinations have been investigated using Amanita muscaria var. persicina as the test organism. The macronutrients and micronutrients found most essential have been supplied as KH_2PO_4 -1.0g, $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ -0.5g, FeCl_3 -0.2mg, ZnCl_2 -0.2mg, and MnCl_2 -0.1mg per liter of medium. Using this inorganic base the following carbon and nitrogen sources have been examined in different combinations: dextrose, sucrose, galactose, fructose, lactose, ribose, maltose, fumaric acid, pyruvic acid, citric acid, malic acid, L-glutamic acid, L-alanine, L-methionine, L-glycine, L-glutamine, L-arginine, L-histidine, L-tryptophan, and L-asparagine. The best growth has been obtained using a medium composed of the basal salts, dextrose and asparagine or glutamine or arginine or histidine

JOHN P. JONES*, Department of Plant Pathology and Physiology, Louisiana State University Agricultural Experiment Station, Baton Rouge, LA 70803. Cytology of spore production in Calonectria crotalariae.

Calonectria crotalariae is an important pathogen on peanuts and potentially important on soybeans in Louisiana. The cytological events associated with conidium and ascospore production were studied as a preliminary to genetic investigations. Conidia receive one nucleus from the conidiogenous

cell; two mitotic divisions occur, so that each cell of the mature four celled conidium contains a single nucleus. The conidiogenous cell becomes, and remains, anucleate when it contributes its nucleus to the developing conidium. The ascus nucleus undergoes typical meiosis and mitosis to produce eight nuclei which are then walled off to form eight ascospores. The ascospore nucleus divides to give the ascospore two nuclei, separated by a septum.

SAEED R. KHAN, Department of Botany
University of Florida, Gainesville,
Florida 32611.

Ultrastructure of septal pore apparatus in the lamellae of Stropharia thrausta

There are three major zones in the lamellae of Stropharia thrausta, an outer hymenium of basidia and scattered cystidia, a subhymenium of interwoven hyphae, and a central trama of parallel hyphae. The septal pore apparatus, throughout the lamellae, consists of a dolipore and associated perforate pore caps. Pore caps are bounded by a unit membrane and have an additional unit membrane in their lumens. Dolipore orifices contain either solid or ring-shaped occlusions. Pore canals are also occluded by narrow zones of electron-dense material. Microfilaments are often present in the pore canals. An extra cap is present on the outside of the pore caps in septa between the subhymenial cells. The septa between tramal cells do not appear to have such outer caps while septa between basidia and subhymenial cells have outer caps only on subhymenial sides. Outer cap appears to be fibrous and is usually bounded externally by cisternae of endoplasmic reticulum or less often by tonoplast of the adjoining vacuoles. The outer cap is visible only after aldehyde fixation and is absent in the potassium permanganate fixed material.

SAEED R. KHAN* AND JAMES W. KIMBROUGH
Department of Botany, University of
Florida, Gainesville, FL 32611. Septal
Ultrastructure in the Tremellaceae

The fine structure of hyphal septa in the Tremellaceae show a number of modifications of organelles found in other basidiomycetes. These appear to be of phylogenetic importance and systems of classification have been proposed based upon septal structure in a limited number of species. This paper presents the fine structure of the tremellaceous genera Basidiodendron, Exidiopsis, Sebacina, and Tremellodendron. All species examined have dome-shaped pore caps bounded by unit membrane and with an electron dense lamella in the pore cap lumen. Pore caps are non-perforate in Tremellodendron and Basidiodendron, while narrow, discontinuous or distinct breaks are found in Exidiopsis and Sebacina. These features are common in Exidia and other heterobasidiomycetes with the exception of Tremella and members of the Filobasidiaceae. It is suggested that schemes of classification based upon the fine structure of a limited number of species may be premature.

JAMES W. KIMBROUGH

Department of Botany, University of Florida
Gainesville, FL 32611. New Species of
Termitaria (Termitariales, Deuteromycetes) from
Australian termites.

Species of Termitaria are external parasites of many genera of termites, forming a variety of lesions on the exoskeletons and sending haustoria into the integument. Each thallus resembles an apothecium in having a discoid hymenium bordered by a sterile, pigmented excipulum. The hymenial area, however is composed of closely adhering phialids, and the basal layer of the excipulum contains darkly pigmented haustorial mother cells. Three species of Termitaria have been described thusfar, occurring on eight termite genera. This report consists of the description of three undescribed species of Termitaria infecting species of the previous unreported host genera Coptotermes, Prototermes, and Mastotermes, and the extension of the host range for T. coronata to the genus Nasutitermes. The new taxa of Termitaria differ from previously described ones in the number of spores per phialid, size and shape of spores, pigmentation of the conidiogeneous zone, size and distribution of haustorial mother cells, morphology and branching of haustoria, and thallus dimensions.

Dennis J. Kitz*, Robert W. Embree and John Cazin, Jr. Departments of Botany and Microbiology, Univ. of Iowa, Iowa City, Iowa 52242.
The Range in Virulence for Six Strains of Absidia corymbifera for Mice.

Absidia corymbifera is a known etiologic agent of mucormycosis, and has been reported from both human beings and lower animals. The virulence of six strains of this fungus was tested by administration of graded spore dosages intravenously and intraperitoneally to normal and cortisone-treated Swiss mice. Mice challenged intravenously received from 10^2 - 10^5 spores. Infections in these mice involved the brain and kidneys, and at higher dosages occasionally the heart and lungs. Doses from 10^5 - 10^7 spores were administered to mice challenged intraperitoneally. Infections in these animals involved the kidneys, and often the spleen and liver. Comparison of LD₅₀ values from these experiments revealed a range in virulence for the six strains of A. corymbifera tested.

KOEHN, ROBERT D.
Department of Biology
Southwest Texas State University
San Marcos, Texas 78666

Stromatal ontogeny in Poronia oedipus

This project observed the origin and development of stromata of Poronia species. In nature it appears that the bulbous bases of stromata are rooted in the dung substrate with an apparent initiation in the substrate. This study observed initiation of stromata with the use of scanning electron microscopy in agar cultures. Using SEM techniques and 5-8 day cultures of P. oedipus it was noted that development begins with the forma-

tion of hyphal aggregates or "knots" on the mycelial surface. Proliferation of hyphal elements follows the appearance of the "knots." The hyphal elements then orient themselves in a vertical manner with the outer filaments forming an ectostroma and inner elements coalescing to form the central shaft. Under laboratory conditions there is no evidence of subsurface growth to anchor the stromata.

JAMES L. KOEVENIG* and E. LIU. Department of Biological Sciences, University of Central Florida, Orlando, Florida 32816 and Department of Biology, University of South Carolina, Columbia, S.C. 29208. - Carboxymethyl cellulase activity in the Myxomycete Physarum polycephalum.

Carboxymethyl cellulase (CMCase) activity was measured viscometrically in homogenized axenic cultures of Physarum polycephalum incubated with CMC. Protein concentration was determined using the BioRad assay. Activity ranged from 140 CMCase units/mg protein/hr in old "orange" plasmodia to 2425 CMCase units/mg protein/hr in dry spores from mature sporangia. Activity in sclerotia and in fruiting plasmodia was higher than in vigorous "yellow" plasmodia. Previous reports of cellulose-like compounds in the spores and fruiting bodies lead one to expect cellulase activity in Myxomycetes, with highest activity in spores and fruiting plasmodia. Ecological implications of CMCase activity are discussed.

GARY F. LEATHAM*, MARK A. STAHMANN, Biochemistry Department, University of Wisconsin, Madison, Wisconsin 53706.--A Fruit-Inducing Substance for the Edible Japanese Black Forest Mushroom Lentinus edodes (Shiitake) (Berk., Sing.).

An extract of oak bark was found to induce the formation of fruiting bodies in pure cultures of Lentinus edodes, which did not produce fruiting bodies without the addition of the extract. With this as the basis for a bioassay, in flasks containing a slurry culture; purification of the fruit-inducing substance(s) is being conducted and its physical and chemical properties studied. These studies have shown that the formation of fruiting bodies may require: (1) A compound that chelates trace elements; (2) Certain trace elements; and (3) An unknown presumably organic factor. When testing known compounds, some metal preparations were active at the parts-per-billion level in the presence of known natural chelators.

LILLY, WALT W.* AND IRIS CHARVAT. Department of Botany, 220 Biological Science Center, University of Minnesota, St. Paul, MN 55108. Acid Phosphatase in Schizophyllum commune; Regulation by Inorganic Phosphate. Acid phosphatase specific activities and isozyme patterns have been examined in extracts from wild-type homokaryotic and dikaryotic mycelia grown for 4 days on membrane-agar plates in which the in-

organic phosphate concentrations of the media were varied. Specific activity of acid phosphatase from homokaryotic hyphae increases linearly versus the $-\log (P_i)$ in the growth medium. One major band of activity is resolved by disc gel electrophoresis when the mycelia have been grown on 9.2mM (minimal medium), 5mM and 1mM P_i media. However, extracts from homokaryotic mycelia grown on P_i concentrations of 500uM, 100uM, 50uM, and 10uM show the same major band plus a second band of somewhat lower activity. The intensity with which this additional band stains appears to be dependant upon the media (P_i). In contrast to the homokaryon, acid phosphatase activity in extracts of colonies of the dikaryon remains essentially constant over the range of P_i concentrations from 100mM to 10uM in the medium. However, the extracts show isozyme patterns similar to the homokaryon with the appearance of the second band at either 500uM or 100uM P_i . This data may indicate that the homokaryon and dikaryon regulate acid phosphatase activity differently.

FRANK L. LYON*, CHARLES L. KRAMER, Division of Biology, and MERLE G. EVERSMEYER, USDA SEA, Dept. of Plant Pathology, Kansas State University, Manhattan, KS 66506.

Airspora profiles: A three year survey.

For the past three years we have investigated the airspora in Manhattan, Kansas and surrounding areas. The objectives of this research has been: 1) to examine the homogeneity of airspora in the atmosphere within and between a variety of ecological sites and 2) to study the relationship between several climatological measurements and numbers of airspora using multiple regression analysis. All sampling was done using Kramer-Collins 7-day drum samplers. This is a suction type sampler which can be used to quantify the numbers of airspora/m³ of air. This paper will present data on airspora profiles from locations ranging from urban sites in Manhattan, KS to sites located in the tall grass prairie areas surrounding Manhattan.

E. S. LUTTRELL. Dept. of Plant Pathology, University of Georgia, Athens, GA 30602. -- Development of Phyllachora lespedezae.

Phyllachora lespedezae produces in living leaves of Lespedeza stipulacea two types of ascocarp: single perithecia, which quickly produce ascospores and substitute for a conidial stage, and perithecia developing in groups within stromata, which slowly mature in dead overwintering leaves. Perithecia from infections on immature leaves develop on fully expanded leaves in 2 weeks, discharge ascospores for 3 weeks, and deteriorate in dying leaves. The mycelium is intracellular. Perithecial initials originate within a single host cell whose cell wall is dissolved. Host cells are progressively destroyed as they are replaced by tissues of the expanding perithecium, but immediately adjacent host cells remain alive. The centrum is made up of paraphyses and asci. The ostiole is periphysate. Stromata of compound ascocarps in fallen leaves contain only young perithecia. These produce ascospores in the following spring.

CHARLES W. MIMS. Dept. of Biology, S.F. Austin State Univ., Nacogdoches, Texas 75962. Ultrastructure of Spore Formation in the Caecoid Telial Stage of the Rust Kunkelia nitens.

The rust Kunkelia nitens infects leaves of the common dewberry Rubus trivialis. The lower epidermis of an infected leaf is eventually ruptured by the fungus revealing a mass of bright yellow to orange spores. These spores are morphologically identical to aeciospores of the rust Gymnoconia peckiana, but are functionally teliospores since upon germination they form basidia and basidiospores.

The ultrastructural aspects of teliospore formation in K. nitens is basically similar to aeciospore formation in those rusts examined thus far. Sporogenous cells give rise to binucleate spore initials in a basipetal fashion. Each initial is produced from the open end of a sporogenous cell and delimited by a centripetally developing septum. The nuclei within the spore initial then divide conjugately and an unequal cell division results in the formation of a small binucleate disjunct cell and a large teliospore. The disjunct cell dies while the teliospore continues to develop. The cytoplasm of the spore becomes very dense and large numbers of lipid droplets appear. Small electron-transparent deposits appear on the outer surface of the plasma membrane of the spore just beneath the primary wall. These deposits develop into spines as a result of the deposition of material beneath them. During this time an apparent gelatinization of the primary wall takes place. The surface of a fully formed spore is covered with these spines.

CHARLES W. MIMS* and E. LAURENCE THURSTON. Dept. of Biology, S.F. Austin State Univ., Nacogdoches, Texas 75962, and Dept. of Biology, Texas A&M Univ., College Station, Texas 77843. Ultrastructure of Teliospore Formation in the Rust Puccinia podophylli.

Puccinia podophylli is an autoecious rust parasitizing Podophyllum peltatum. Infected leaves of the host bearing telia of the fungus were collected and prepared for study with both TEM and SEM.

Teliospore initials of P. podophylli develop from binucleate sporogenous cells lining the base of the telium. The nuclei of the sporogenous cell divide conjugately and two of the daughter nuclei move into the teliospore initial while two remain in the sporogenous cell. The initial elongates and is delimited from the sporogenous cell by the formation of a septum. The nuclei within the initial then divide conjugately and a septum develops separating the initial into two binucleate cells. The basal cell becomes the pedicel of the spore while the apical cell continues to develop into a two-celled teliospore.

Mature teliospores of P. podophylli are oblong or clavate and are sparingly covered with rather long straight or slightly curved spines. These spines initially appear as slight bulges on the surface of the young spore. The spore wall in such a region then evaginates to form a slender projection filled with cytoplasm. A central

electron-translucent lumen devoid of cytoplasm is visible in older spines viewed in thin section. Little substructure is evident in the extremely electron-opaque wall of older teliospores.

DAVID J. MCLAUGHLIN, Dept. of Botany, University of Minnesota, St. Paul, MN. 55108--Septation and Maturation of Auricularia Basidia.

Septum formation and vacuolation in the phragmobasidia of Auricularia polytricha were examined using light microscopy and transmission electron microscopy to determine the manner of basidial development. Septa within basidia develop centripetally as in hyphae and form a septal pore apparatus. The latter differs from those at the basidial base in size of the pore and pore cap. Basidial compartments are evacuated in basipetal sequence. As each segment is evacuated the septal pore apparatus breaks down and the pore is sealed. A large volume of cytoplasm appears to enter each compartment during sterigma (protosterigma) growth and a basal vacuole(s) enlarges late in sterigma or early in spore formation. The pattern of development is thus far unique among organisms with phragmobasidia and differs significantly from that of typical holobasidia.

Perry Mülleavy* and O'Neil Ray Collins. Department of Botany, University of California, Berkeley, CA 94720. Mating-type heterozygosity: effects on mating in diploid myxamoebae in the myxomycete Didymium iridis.

Experiments were designed to determine whether mating-type myxamoebal heterozygotes (A^7A^8) of Didymium iridis are capable of mating. Ploidy levels in all experiments were determined by measuring nuclear DNA content by means of a scanning microspectrophotometer. Two haploid myxamoebal color mutants (A^7b^2 and A^8b^2) were diploidized by treatment with isopropyl-N(3-chlorophenyl)carbamate (CIPC). The diploid mutants were crossed and produced a $4N$, cream-colored plasmodium (wild-type color = brown). Six of 23 F_1 progeny from this plasmodium did not form plasmodia in single-spore culture, whereas the remaining 17 did. The 17 diploid apomictic clones were heterozygous for mating type and produced cream-colored plasmodia. This was expected since the tetraploid plasmodia possessed only recessive color alleles ($b^2b^2b^2b^2$). Five of these facultative apomictic F_1 clones were crossed with CR2-25, a $2N$ mating-type homozygote (A^2A^2) which carries wild-type color alleles (b^2+b^2). In three of the 5 crosses, both brown and cream plasmodia developed. The $4N$ brown-colored plasmodia were the result of mating between the two clones, whereas the $2N$ cream-colored plasmodia developed apogamically. The F_2 generation derived from a brown-colored plasmodium is being studied to determine meiotic segregation ratios of mating type and color alleles.

PALM, M. E.* and E. L. STEWART. Department of Plant Pathology, University of Minnesota, St. Paul, MN 55108. The Genus Trichocladium (Deuteromycotina, Hyphomycetes).

Trichocladium Harz was described in 1871 for dematiaceous Hyphomycetes with fragile hyphae which produced solitary to clustered, thick-walled, oval to ovoid blastospores, with one to several transverse septa, on poorly differentiated conidiophores. Since then the generic concept has been expanded to include dematiaceous Hyphomycetes which have conidiophores that are absent or micronematous to semi-macronematous, mononematous, scattered or forming sporodochia. Conidiogenous cells are holoblastic, monoblastic or occasionally polyblastic, integrated, terminal and intercalary, determinate, cylindric, doliform, subglobose, or tapering. Conidia are solitary, acrogenous, acroporeogenous, or pleurogenous, dry, with one to many transverse septa, sometimes constricted at the septa, pigmented, usually thick-walled, smooth, roughened or lobed, cylindric, clavate, ovoid, obovoid to pyriform to subglobose, with or without germ pores. Accessory spore states have been recorded. Twenty-two Trichocladium species exist in the literature. Thirteen Trichocladium species are currently accepted. The genus, as now circumscribed, includes a fairly heterogeneous group of species, but is being maintained at present. Germ pores in conidia and accessory phialidic states are being evaluated as subgeneric criteria in this study.

RONALD H. PETERSEN.* Department of Botany, University of Tennessee, Knoxville, TN 37916.

The dismemberment of the Schweinitz collections. Fungi from Schweinitz's herbarium were given to Torrey, Curtis, Berkeley and Fries, and now may be found in several major collections. The details of these transfers are traced through letters by Schweinitz, Torrey, Curtis and Berkeley.

J. POMMERVILLE and W. BARSTOW* Department of Botany, University of Georgia, Athens, GA 30602. The Cytology of Cell Wall Formation during Mitospore Development in Allomyces. The formation of a cell wall was followed by light and electron microscopy on synchronously germinating populations of A. macrogynous mitospores. Using the fluorescent brightener Calcafluor, which will react with chitin, a cell wall could not be detected around the developing mitospore until 5-6 min after the induction of germination. Ultrastructural analysis of the developing cells showed the following: (1) 1 to 3 min after induction the cytoplasm became dense near the periphery of the cell and small vesicles 35-70 nm diam were concentrated in this area; (2) by 4 to 6 min some of the vesicles were fusing with the plasma membrane, releasing their contents into the extracellular medium and initiating the formation of a cell wall; (3) the vesicles that gave rise to the initial cell wall appeared to arise from the γ -particles found in the cytoplasm of the mitospore; (4) the nuclear cap and retracted flagellar axonemes did not break down until after the wall had been formed, ca. 10-15 min after induction. These results indicate that the initial response of the mitospores to induction is the formation and release of small vesicles from the γ -particles. The developmental fate of the γ -particles of Allomyces appears similar to that of Blasto-

cladiella emersonii [Truesdell and Cantino, Arch. Microbiol., 70:378-392 (1970)]. We suggest that the γ -vesicles of Allomyces may be the functional equivalent of chitosomes [Bracker, et al., Proc. Natl. Acad. Sci. USA, 73:4570-4574 (1976)].

POPE, LEODOCIA M. AND GARRY T. COLE.*
Departments of Microbiology and Botany,
University of Texas, Austin, Tx. 78712.
Experimental candidiasis in the infant
mouse.

Infant mice can be chronically or lethally infected by intragastric challenge with Candida albicans. No prior manipulation designed to compromise the host is performed except for a 5 hour period of fasting just before challenge. The yeast administered to the mice quickly become systemic in distribution and extensively colonize the digestive tract. Scanning electron microscopic examinations of C. albicans in situ reveal variations in the surface texture of the yeast in different parts of the gastrointestinal tract. The surface ultrastructure of yeast extracted from various parts of the gastrointestinal tract are investigated. In addition, germ tube formation appears to be most frequent in the distal portion of the tract (i.e. caecum and large bowel). Results of thin section examination of these yeast at different stages of germination are presented. The possible significance of these data in colonization and infection is discussed.

MARTHA J. POWELL. Botany Department, Miami University, Oxford, Ohio 45056. - Ultrastructural identification of chromidia in the aquatic fungus Polyphagus euglenae.

In classical studies of Polyphagus euglenae Nowakowski, a chytrid parasite of Euglena, densely granular networks surrounding nuclei were interpreted as chromatin extruded from the nucleus and used for lipid synthesis. These networks were called chromidia. Zoospores, prosperangia, and asexual sporangia were studied with electron microscopy to determine the ultrastructural identification of chromidia. In the zoospore the nucleus was enclosed by an aggregation of ribosomes. In young prosperangia a network of microbody-like structures surrounded nuclei. In more mature prosperangia, microbody-like structures associated with lipid globules were concentrated around the single, large primary nucleus. When the primary nucleus migrated into the asexual sporangium, the microbody-like structures still surrounded the nucleus; but after the sporangium became multinucleate, microbodies were scattered throughout the cytoplasm. When incubated in the diaminobenzidine medium for the cytochemical detection of catalase, reaction product was found in these microbody-like structures, confirming that "chromidia" described in prosperangia and asexual sporangia by classical mycologists are really microbodies. Rather than giving rise to lipid, these microbodies are probably involved in the metabolism of the lipid globules with which they are associated. The "chromidia" in zoospores are not extruded chromatin as suggested earlier, but correspond in their location around the nucleus to an aggregation of ribosomes.

ROBERT M. QUEEN* and JAMES L. KOEVENIG.
Department of Biological Sciences, University of Central Florida, Orlando, Florida 32816.
- Immunological studies of mating strains of Myxomycetes.

Immunological differences between clones of several mating strains of the Myxomycetes Didymirum iridis and Physarum polycephalum, grown on E. coli, were analysed using fluorescent antibody staining, gel diffusion and immunoelectrophoresis. Results show no significant differences between mating strains for a species. Absorption of surface proteins had no significant effect on plasmodial development.

IAN. D. REID. National Research Council of Canada, Prairie Regional Laboratory, Saskatoon, Sask., Canada S7N 0W9. -- Conjugation hormones in Fibulobasidium inconspicuum.

Fibulobasidium inconspicuum Bandoni is a heterobasidiomycete with a haploid yeast phase and a tetrapolar mating system. In compatible crosses the yeast cells produce filamentous conjugation tubes. These tubes are induced by dialysable hormones. Unmated cultures do not normally produce detectable amounts of hormone; they must first be stimulated by an inducing factor from the other mating type.

ALVIN L. ROGERS*. Departments of Botany and Plant Pathology and Microbiology and Public Health, Michigan State University, East Lansing, Michigan 48824. Medically Important Fungi in the Introductory Mycology Course -- Ascomycetes and Basidiomycetes.

The perfect stages in the life cycle of many fungi that are pathogenic in humans and animals have been discovered during the last 20 years. Most of the investigations resulted in the organisms being placed into the family, Gymnoascaceae, in the Euscomycetes. Many of these fungi are soil and/or manure inhabiting organisms in their natural saprophytic existence. Among the common medically important fungi only the perfect states of Cryptococcus neoformans which cause cryptococcosis, Filobasidiella neoformans and F. bacillospora have been found to be basidiomycetes. These two organisms are too dangerous to be used as live material in experiments for an Introductory Mycology Laboratory. But a number of the ascomycetes that are in the soil which very rarely cause ringworm (tinea) in humans and animals may be used in experiments to illustrate certain phenomena in the Kingdom of Fungi. The life cycle of the ascomycete Nannizzia incurvata (Microsporum gypseum) and/or N. fulva (M. fulvum) which can be easily isolated from many garden soils using sterilized horse hair may be used to study the following:

- The imperfect and perfect stages in a typical life cycle (of the Euortiales).
- Soil as a habitat for fungi.
- Selective isolation using hair as a substrate.
- Formation of macronidia and micronidia and their use in the classification of the Fungi Imperfecti.
- etc.

ROSS,* L., B. ENGLE, E. BUNDERSON, D. J. WEBER and G. BOOTH. Dept. of Botany and Range Science, Brigham Young University, Provo, Utah 84602. Toxicity and physiological changes in fungi inhibited by triphenyltin hydroxide.

The LD₅₀ for triphenyltin hydroxide (TPTH) was 2 ppm for *Rhizopus arrhizus*, *Aspergillus niger*, *Fusarium oxysporum*, *Pythium debaryanum* and *Trichoderma viride* using an agar growth medium. In liquid culture, growth of *Aspergillus niger* was inhibited 50% at 0.01 ppm. The number of soil inhabiting fungi were determined in three soils (clay, loam, and sand) that had been treated with 1 ppm of TPTH. There was no significant difference between the number of colonies in the control and the 1 ppm TPTH-treated clay soil. There was a significant difference between the number of colonies in the 1 ppm TPTH treated and the control loam and sandy soils. TPTH (10 ppm) had no significant effect on the respiration of *Aspergillus niger* as measured by the Warburg apparatus. TPTH (100 ppm) was non-inhibitory to alpha amylase, cellulase, and pectinase.

C.A. Shearer* and A.D. Hewings. Dept. of Botany, Univ. of Ill., Urbana, IL 61801. Quantitative Analysis of the Community Structure of Aquatic Hyphomycetes on Submerged Leaves.

The structure of a community of aquatic hyphomycetes in a midwestern river was analyzed by measuring frequency of occurrence of individual species in samples from 100 leaves selected randomly. Ten discs were cut from each leaf, incubated in dilute salt at stream temperature for 3 days, then fixed, stained and examined for conidia. The number of observations of a species on single leaves divided by the number of observations of all species on single leaves was used to estimate the relative ability of species to disperse and colonize substrates. The number of observations of a species on all discs divided by the number of observations of all species on all discs was used to estimate the relative ability of species to competitively colonize leaf area. The sum of these two measurements was used as a relative importance value for each species. Importance values were used to calculate Shannon-Weaver diversity (H') and evenness (J). In a hyphomycete community sampled on Nov. 13, 1977, species diversity (D) was 17, H' was 3.0 and J was .73. The predominant species in order of relative importance were *Tetracladium marchalianum*, *Filosporella annelidica*, *Clavariopsis aquatica*, *Lemonnieria aquatica*, *Anguillospora longissima* and *Alatospora* "complex". The min., max. and av. number of species per disc was similar for silver maple, sugar maple, oak and sycamore leaves and ranged from 0-8 with an overall mean of 3.51. Definite positive and negative associations between species were not observed. The dominant species, *T. marchalianum*, co-occurred with every other species and the next three dominant species occurred with all other species except *Tricladium angulatum*, *Flagellospora curvula* and *Triscelophorus monosporus*, three species which appeared very rarely in this collection.

MARTHA A. SHERWOOD*, Farlow Herbarium, Harvard University, Cambridge, MA 02138. ELIZABETH J. KNEIPER, Winsor School, Boston MA 02215. The former and present lichen flora of the Boston metropolitan area.

Extensive lichenological collections were made in the Boston metropolitan area in the 19th century. A comparison of these early records with the present flora indicates that the changes wrought by the combined effects of pollution and habitat destruction have been profound. Our total of 81 species compares favorably in diversity with that reported by 19th century lichenologists, but includes a number of species which have either invaded the area in recent years or increased markedly in abundance. Sensitivities of species to air pollution, as deduced from their distribution records, are similar to those observed on Long Island and in Europe.

R. D. SLOCUM. Cell Research Institute and Department of Botany, University of Texas, Austin, TX 78712. Characterization of Concentric Bodies From Ascomycetous Fungi.

Concentric bodies are morphologically unique, subcellular particles found in numerous representatives of both lichenized and non-lichenized ascomycetous fungi. We have been investigating the morphological and biochemical attributes of these bodies in order to elucidate their, as yet, unknown functional nature in the fungal cell. Current studies involve the physical and chemical characterization of concentric bodies both *in vivo* and isolated and purified from fractionated mycelia using sucrose density gradient centrifugation, SDS polyacrylamide gel electrophoresis and electron microscopical techniques. Preliminary data will be presented in view of the suggestions that concentric bodies may represent either a previously undescribed type of eucaryotic organelle or a new class of mycoviruses.

FREDERICK W. SPIEGEL, Department of Biology, Princeton University, Princeton, N.J. 08544. Cellular reorganization during culmination of Polysphondylium pallidum.

During culmination the sorogen of the dictyostelid cellular slime mold Polysphondylium pallidum deposits rings of amoebae formed at its posterior end at regular intervals along its stalk. Each ring of cells breaks up into one or more secondary sorogens so that a whorl of small, Dictyostelium-like sori is produced at maturity. Light microscopy shows that a ring is first visible as a slight swelling of the posterior portion of the sorogen. The ring decreases in length and increases in diameter and ultimately separates from the primary sorogen. Electron

microscopy of thinsections of sorogens shows that cells of the ring orient toward the edge of the ring in a manner homologous to that shown by cells at the stalk-forming tip of the sorogen. Thus, it is postulated that certain cells in the posterior region of the sorogen become tip-like causing other cells nearby to reorient and move toward them rather than toward the tip.

W. L. STEFFENS and J. P. JONES. Department of Plant Pathology and Crop Physiology, Louisiana State University, Baton Rouge, LA. 70803.---The Ultrastructure of Ascospore Orientation and Delimitation in Eleutherascus peruvianus.

Scanning and transmission electron microscopy of samples at representative developmental stages, has revealed some unusual features in relation to ascospore ontogeny in this simple Discomycete. A persistent symmetrical arrangement of the ascospores in relation to spore wall ornamentation suggests that the ascospores are spatially defined within the asci rather than randomly arranged, up to a certain period of development. This is demonstrated by stereo scanning electron microscopy of sectioned asci. Correlative studies, utilizing transmission electron microscopy, have revealed prospore delimitation by ascus vesicle formation. Events observed during early delimitation and spore wall secretion, suggest that spore nuclear positioning, delimitation, and ultimately the formation of outer wall ornamentation may involve a complex radiating array of cytoplasmic microtubules operating in a mechanism proposed in part by several earlier investigators.

ELLIS E. SYKES* and DAVID PORTER, Department of Botany, University of Georgia, Athens, Georgia 30602. Resistant Sporangia Development and Early Infection in Catenaria allomycis (Blastocladales).

Catenaria allomycis is an obligate parasite of species of Allomyces. The development of resistant sporangia has been investigated using light and electron microscopy. Synaptonemal complexes have been identified in young resistant sporangia. This observation suggests that the primary spores that are derived from the resistant sporangia are haploid. Upon release from the sporangium, the primary spores encyst within a short time near the discharge tube. Within 3 to 4 hours, the encysted spores form papillae and the cytoplasm cleaves into four gametes which are soon released from the cyst. Gamete fusion produces zygotes which infect host hyphae in a manner similar to that observed for zoospores. Infection of Allomyces hyphae by the C. allomycis zoospores can conveniently be divided into 3 phases: (1) the formation of a germ tube which is apically appressed to the host wall, (2) the production of a penetration tube which actually pushes into the host but does not rupture the host plasma membrane, and (3) the rupture of the host plasma membrane to release the parasite directly into the cytoplasm.

CARL B. WOLFE, JR.
Biology Department
Pennsylvania State University
Mont Alto, PA 17237

A New Genus of the Boletaceae

Type studies of the various taxa assigned to Porphyrellus revealed a small but discrete group of taxa quite dissimilar to P. porphyrosporus Fr. (type species of Porphyrellus). This group of taxa could not be easily included in any of the genera of the Boletaceae or Strobilomycetaceae as summarized by Singer (1975). Consequently, the recognition of a new genus to include these taxa became necessary. This genus differs from Porphyrellus in three major respects. First, the trichodermium of the new genus is an ixotrichodermium and not a palisade trichodermium as in Porphyrellus. Secondly, the pleurocystidia of the new genus taxa are void of microchemically positive contents; pleurocystidia in Porphyrellus have microchemically positive contents. Thirdly, the spore E^m value (length/width ratio) of the new genus is 3.0; in Porphyrellus the spore E^m value is always less than 3.0. The recognition of this new genus seems justified on the basis of the major differences in these three character states.

J.W. TAYLOR* and M.S. FULLER Department of Botany, University of Georgia, Athens, GA 30602. Zoosporogenesis and the apophysis-sporangium interface in Chytridium confervae.

Light and electron microscopic observations have shown that the encysted zoospore of Chytridium confervae (Wille) Minden (monocentric, eucarpic, Chytridiales) develops into a sporangium with an apophysis and rhizoids. During growth, the cytoplasm is continuous between the multinucleate sporangium and the anucleate apophysis and rhizoids. When zoosporogenesis is induced by starvation the cytoplasmic continuity changes. First, an array of many fibers (0.5 μ m x 30.0 μ m) forms between the apophysis and sporangium. Microtubules are one component of these fibers. At the junction between the apophysis and sporangium, where the density of fibers is greatest, there is a sharp interface between the ribosome-rich cytoplasm of the sporangium and the ribosome-poor cytoplasm of the apophysis. Second, after approximately two hours a septum forms centripetally between the apophysis and sporangium. The sporangium is then isolated from the apophysis, except for plasmodesmata in the septum. We speculate that the fibers are involved in concentrating the cytoplasm and ergastic substances in the sporangium and perhaps also in orienting the developing septum. Plasmodesmata in the septum suggest that communication between the sporangium and apophysis is necessary during zoosporogenesis.

THE MYCOLOGICAL SOCIETY OF AMERICA

Application for Membership

Name _____

Mailing address _____

_____ ZIP _____

Do you desire to be ☐ Regular, ☐ Student, or ☐ Associate Member?

Date on which you wish your membership to begin: January 1, 19____.

Signature of a member endorsing your application _____

Any person interested in the study of fungi is eligible for membership. Annual dues of Regular Members are \$15.00. A special annual rate of \$8.00, for a maximum of 5 years, is provided for a member enrolled as a student. Regular and student members receive MYCOLOGIA and the MSA NEWSLETTER, may participate fully in the annual meeting, may submit papers for publication in MYCOLOGIA, and may vote and hold office in the Society.

Annual dues for Associate Members are \$5.00. These members receive the MSA NEWSLETTER and may present one paper at the annual meeting of the Society. They do not receive MYCOLOGIA or have other privileges of membership (Information about Life Membership may be obtained from the Secretary-Treasurer.)

Membership may begin either on January 1 of the current year or on next January 1.

Mail this application along with a check, money order, or bank draft in U.S. Funds covering the first year's dues to:

Dr. Harry D. Thiers
Dept. of Biology
San Francisco State University
1600 Holloway Ave
San Francisco, CA 94132 USA